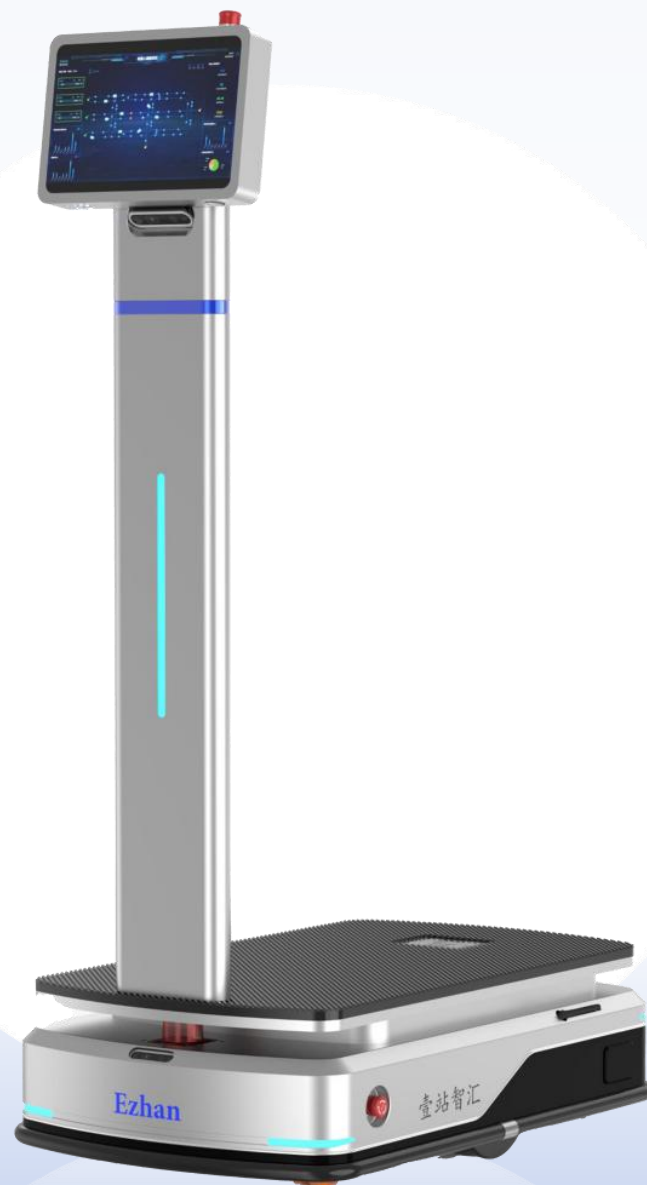
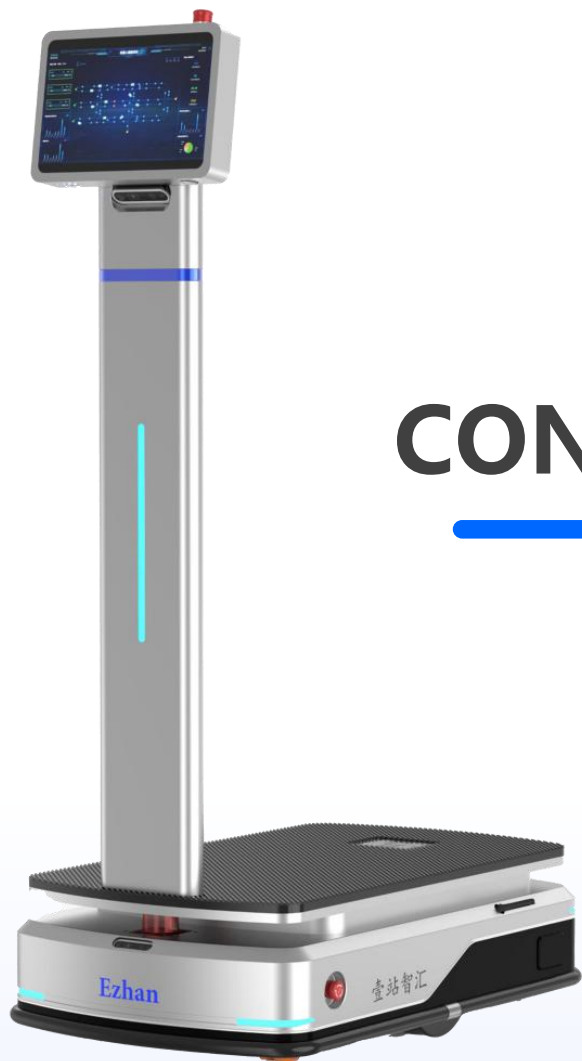




E300 Series

User Manual



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- PART 08 Routine Maintenance**
- PART 09 AMR Usage Guidelines**
- PART 10 HCI Platform Instructions**

1

PART ONE

Product Introduction

- Product Profile
- Main Purpose and Applicability
- Disclaimer
- Operations and Environment Conditions

○ Product Profile

The E300 series is a mobile robot based on laser SLAM navigation.

Advantage

- It autonomously plans optimal paths to complete tasks efficiently and has a excellent environmental adaptability.
- It is easy to maintain , easy to understand the operation process , and users can quickly get started , which significantly improves the efficiency of automated operations.
- By integrating lidar and 3D camera, it can accurately determine whether there are obstacles around it and automatically detour around them when the vehicle encounter them, ensuring smooth operation.
- In addition, the robot is equipped with a complete safety mechanism, including low battery warning and emergency stop protection and other functions to ensure safe operation.

Model Description

This manual applies only to the E300 series (E300 and E300 Pro).

- The content of this document has been verified and confirmed to be as accurate, reliable, and complete as possible. However, no guarantee is provided that all content aligns precisely with the actual product. Technical specifications are subject to change without prior notice.
- No organization or individual may reproduce or distribute this document in any form without written permission. When referenced, the source must be cited, and no misrepresentation, abridgment, or modification of this document is permitted.
- Shenzhen Ezhan Technology Co., Ltd. reserves the right to amend this disclaimer and holds the final right of interpretation.

○ Main Purpose and Applicability

E300 Series is designed for cargo handling.

- It is widely used in multiple fields including material transfer in workshop production lines, flexible assembly lines, warehouse storage logistics, catering services and medical services.
- With the assistance of the interactive interface and the robot control terminal, the E300 Series Products robots can provide efficient, flexible and intelligent logistics solutions, significantly improving the level of operation efficiency and intelligence.

○ Operation and Environmental Conditions

The E300 Series features intuitive operation

- Simply use the interactive interface to perform laser scanning for real-time map building, path planning, and station setup—ready for immediate deployment.

The E300 Series is capable of auto-locating its station positions

- Independently planning routes, and autonomously navigating around obstacles.

Do not carry liquids when moving items to prevent liquid from entering the machine's internal components, causing malfunctions and damage.

Low concentration of airborne particulate matter (dust, metal particles, etc.), and no continuous vibrations or excessive impacts

No long-time direct exposure to sunlight

Compatibility with the environment: $\geq 70\%$

Indoor environment with no corrosive gases, explosive gases, harmful gases, or liquids

Temperature: 0°C to 45°C
Humidity: $10\% \sim 90\%$, no compression condensation

2

PART TWO

Performance Parameters

- Main Functions
- Key Functions
- Technology Specifications

○ Main Functions

The E300 Series Products can realize automatic driving with loads, through the interactive interface and the robot control terminal send tasks to execute materials transport, and can independently plan routes and obstacles to complete transportation tasks, realize automated transportation, and minimize manual labor.

Basic parameters

Navigation mode	Laser SLAM
Drive mode	Double wheel differential
Dimension(L*W*H)	785*470*1300mm
Weight(with batteries)	100kg
Maximum payload	300kg
Rotating diameter	840mm
Classis ground clearance	25mm
Maximum lifting height(Optional)	55mm+2mm
	0-45°C
Ambient temperature and humidity range	10-90%
	No compression condensation

Battery parameters

Battery type	Ternary lithium
Battery capacity	48V/15AH
Comprehensive battery life	6h
Charging time(10%-80%)	≤1.5h
Charging mode	Manual/Automatic
Cycle life	1000cycles capacity≥80%

Performance parameters

Passability (slope/step/gap)	5°/5mm/20mm
Navigation position accuracy	± 10mm
Navigation angle accuracy	± 1°
Minimum passing width	600mm
Max speed(Adjustable)	≤1.5m/s

Technology Specifications

NO.	Functions	Remarks
1	Linear deceleration parking	Slow down upon arrival or encountering obstacles
2	Start linear reinforcement	Start with an increasing speed
3	Choose a stop to park	Automatic parking at the destination station
4	Low voltage alarm	Prompt to charge in time
5	Low battery automatic charging	Automatically charge at charging stations
6	Automatic obstacle bypass	Automatically bypass when encountering obstacles
7	Emergency cut-off	Emergency Stop Button
8	Laser obstacle avoidance	First protection
9	Mechanical collision avoidance	Second protection
10	Emergency braking	Third protection
11	Voice	Alert pedestrians
12	One-key start	Start simply and quickly

3

PART THREE

AMR Components Introductions

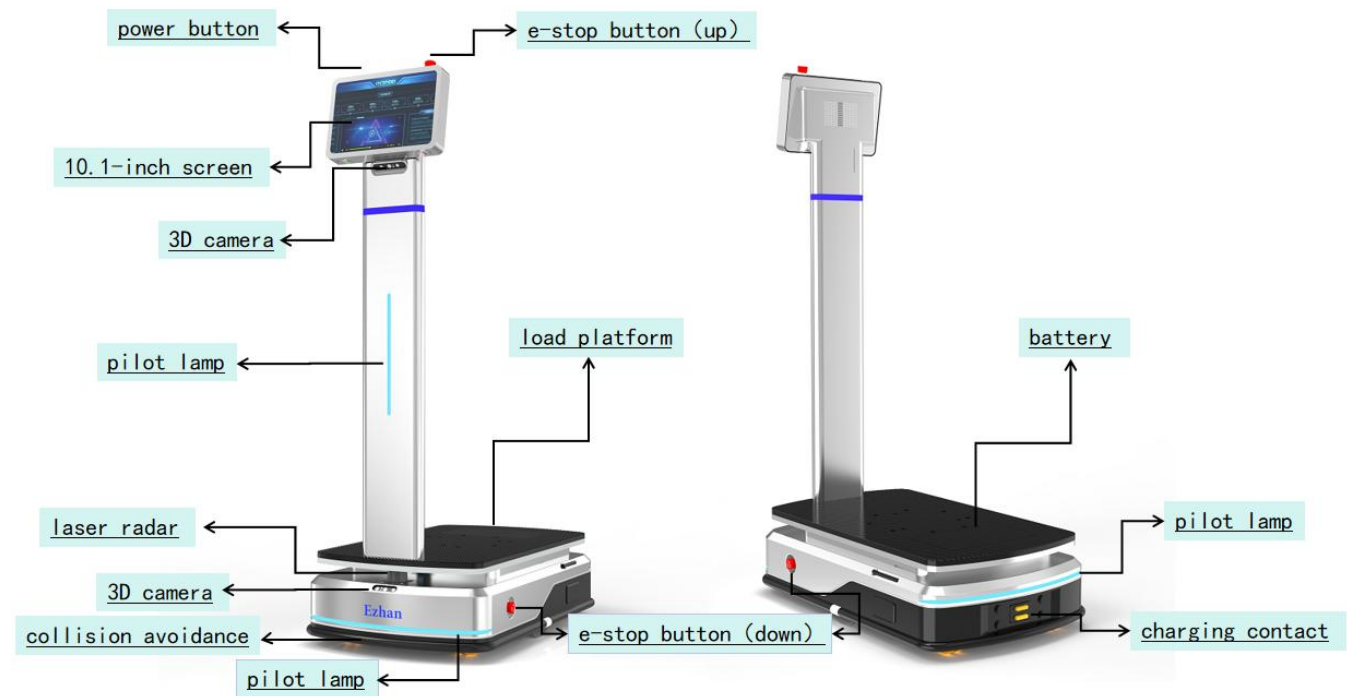
- Introduction of Machine Components
- Button Descriptions for the Control panel

○ Introduction of Machine Components

The orthogonal views depict the E300 Series Products product front elevation and rear aspect with dimensional accuracy.

Instructions for Body Buttons

- **power button:** Press this button to turn the AMR power on or off.
- **e-stop button (up):** Press this button to stop the AMR by decelerating in any situation. In case of an abnormal state, press the emergency stop button and then release it to reset the AMR.
- **e-stop button (down):** Press this button to cut off power to the AMR's drive system, allowing easy manual movement and mapping of the AMR. In case of an abnormal state, press the emergency stop button and then release it to reset the AMR.



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PART FOUR

Laser Navigation

- Detection
- Detection Range

○ Laser Navigation

A sensor used for scanning and mapping, as well as obstacle detection. Its main functions include:

- mapping
- Localization and Navigation
- Obstacle Recognition

Detection

- In front of the robot: The 3D camera and LiDAR are used in conjunction to effectively detect obstacles, with almost no blind spots.
- Both rear sides and the tail: These two positions are blind spots for obstacle avoidance, and obstacles cannot be detected normally. Pedestrians or goods need to be actively avoided.

Detection Range



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PART FIVE

Battery

- Battery System
- Battery Precautions

○ Battery



AMR's battery is designed with a fully sealed and non-polluting structure.



Use a lithium-ion battery, Compared to regular lithium batteries, It offers advantages such as longer range, high energy efficiency, high-power output, and excellent durability.



Its fully sealed design ensures safety and environmental protection, allowing users to achieve optimal performance when used properly.

Battery Parameters

No.	Items	Parameters
1	Rated capacity	15AH
2	Nominal coltage	48V
3	Battery type	Ternary lithium
4	Battery pack voltage range	40.5~54.5V
5	Standard charging current	10A
6	Standard discharge current	10A
7	Operating temperature	-20°C ~+60°C
8	Communication protocol	RS485
9	Cycle life	≥1000 times
10	Battery pack short circuit	Built-in short circuit protection, No explosion, no fire
11	Battery pack overcharge	No explosion, no fire
12	Battery size(L*W*H)	230*125*80mm
13	Battery Weight	4.3kg

Battery Parameters

Items
● Do not leave the battery unused for extended periods. If the battery will not be used for an extended period, disconnect it from the device and fully charge it before storage. It is fully charged every 3 months to maintain its capacity and lifespan.
● Do not store the battery in a sealed container, near open flames, or immerse it in water. Store it in a dry, well-ventilated area away from direct sunlight or heat sources.
● Before using a battery that has been stored for an extended period, ensure it is fully charged and test it for any issues.
● If the battery emits an unusual odor or leaks, immediately stop using it or charging it. Place it in a well-ventilated area away from open flames and notify the after-sales service for handling.
● When connecting the load, ensure correct polarity and avoid reversing the positive and negative terminals.
● Charge the battery promptly when the charge level falls below 5% or when the individual cell voltage drops below 2.8V. Otherwise, the battery may become deeply discharged and unable to charge.
● Charge the battery promptly when the charge level falls below 5% or when the individual cell voltage drops below 2.8V.

Warnings

- Avoid direct sunlight and keep away from open flames to prevent explosions.
- Store the battery in a dry environment, away from rain, leaks, and dusty areas.
- During the battery in a dry environment, away from rain, leaks, and dusty areas.
- When moving the battery, pay special attention to preventing it from falling, which could cause injury.

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PART SIX

Charging Station

- Charging Station Parameters
- Charging Station User Manual
- Charging Station Safety Precautions

○ Charging Station

The E300 Series is equipped with an automatic charging station designed for use when the AMR's battery is running low. AMR can be set to automatically seek out the charging station and initiate charging.

NO.	Items	Parameter
1	Model	E-YLH-10A
2	Dimension(L*W*H)	380*178*330mm
3	Input voltage (AC)	220V~240V
4	Output voltage (DC)	54.75V
5	Output current	10A ± 0.2A
6	Rated power	500W
7	Charging mode	Automatic Alignment
8	Charging type	Contact-based

Key Features

- The charging station is equipped with a high-power battery charger, significantly reducing the charging time for the AMR and ensuring extended operational periods.
- The charger utilizes advanced CPU control and Pulse Width Modulation (PWM) technology, delivering high efficiency, stability, and energy efficiency. It automatically and precisely regulates the charger's charging current and voltage to prevent overcharging, thereby safeguarding the battery's lifespan.

Charging Station Safety Precautions

- Once the charging station's position is determined, it should not be moved or obstructed arbitrarily. Doing so may prevent the AMR from locating and accessing the charging station for recharging.
- The charging station should be placed in a dry, well-ventilated area, away from direct sunlight or heat sources, and must not be positioned near open flames or potential fire hazards.
- Check for the charging icon on the interactive interface when AMR contacts the station. If missing, inspect the station's circuitry or brush block contact.

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PART SEVEN

Abnormal situations and Solutions

- Obstacle Abnormality
- Collision Sensor Abnormality
- Emergency Stop Abnormality

○ Abnormal situations and Solutions

1

The AMR vehicle detects an obstacle in its path

When the AMR detects an obstacle ahead, it will autonomously avoid it. If it cannot bypass the obstacle, human intervention will be required.

2

The Emergency Stop Button on the AMR vehicle has been pressed

Check if there is an obstacle in front of the AMR to trigger the anti-collision switch, if it is caused by an obstacle, clear the obstacle, then pull the button again to release the Emergency Stop, press the Reset Button, and the AMR will start running.

3

The collision sensor device on the AMR vehicle has been triggered

Inspect the area around the AMR's collision sensor to see if it has been compressed or collided with an object. If caused by an obstacle, clear the obstacle, press the reset button to restore normal operation.

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PART EIGHT

Routine Maintenance

Routine Maintenance

- **Precautions of Liquid Cleaning for Laser Sensors:** It is strictly prohibited to clean the laser sensor with any liquid (any damage will not be covered under warranty). If cleaning is necessary, please use a clean, anti-static cloth to gently wipe the sensor, ensuring that no liquid comes into contact with the laser sensor to prevent scratching. Laser sensors are delicate and sensitive components, and any damage resulting from impact or scratching is not covered under warranty.
- **Environmental Consistency for AMR's Path and Stations:** Areas traversed by the AMR and its docking stations must maintain an environmental compatibility level of no less than 70%. If surrounding conditions change significantly, it is necessary to place surrounding items to achieve higher environmental compatibility. Failure to do so may result in the AMR becoming lost or experiencing navigation errors.
- **Stable Placement of Materials:** When placing materials, ensure that they are positioned stably, with a low center of gravity (heavier and larger items should be placed at the bottom of all items). Additionally, ensure that materials will not topple or tilt during transportation by the AMR.
- **Surface Conditions for AMR's Path:** The surface on which the AMR travels should be flat ($\text{Slope} \leq 5^\circ$ / $\text{Step} \leq 5 \text{ mm}$ / $\text{Gap} \leq 20 \text{ mm}$) and dry (without water, cleaning agents, oil, or lubricants) to prevent wheel from slipping and other related issues.

Routine Maintenance

- Ensure that the AMR walking channel is unblocked, do not place obstacles on the walking channel of the AMR. If obstacles are placed on the walking route of the AMR, then AMR will carry out the obstacle bypassing automatically, but when there is a limited room for the AMR to operate a bypassing turning, manual intervention is required.
- Avoid Sudden Approach to AMR: Avoid sudden approaches to the AMR to prevent abrupt stops that may lead to material spillage or other safety issues.
- Surface Treatment and Friction Coefficient: To optimize AMR usage, the areas where the laser AMR operates should not be waxed, and the ground's static friction coefficient should be at least 1.0. Damage to the AMR resulting from wheel slipping on the floor due to waxed surfaces is not covered under warranty.



PART NINE

AMIR Usage Guidelines

○ AMR Usage Guidelines

The AMR's safety obstacle avoidance detection device is located at the front of the vehicle, resulting in certain blind spots during operation. Special attention must be paid to the following situations:

- ① **During Automatic Charging:** When the AMR is docking with the charging station in reverse, it cannot detect obstacles behind it. Ensure that the AMR walking channel is unblocked, do not place obstacles on the walking channel of the AMR.
- ② **During Entering a Rack:** During the process of identifying and entering a rack, the obstacle avoidance function is disabled, and obstacles cannot be detected.
- ③ **During Turning or Rotation:** When the AMR is turning or rotating in place, the rear sides and rear of the vehicle become blind spots, and obstacles cannot be detected.

To ensure smoother and more stable operation of AMR on site, When encountering the above situations, please remind site personnel to actively avoid AMR to prevent collisions and abnormalities. Do not place obstacles on the path. Additionally, strengthen personnel training and awareness campaigns to enhance usage standards.

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PART TEN

HCI Platform Instructions

- Prepare AMR Operation Area
- Use HCI Platform

Notes: HCI Platform - (Human-Computer Interaction Platform)

E300 Series Operation Manual

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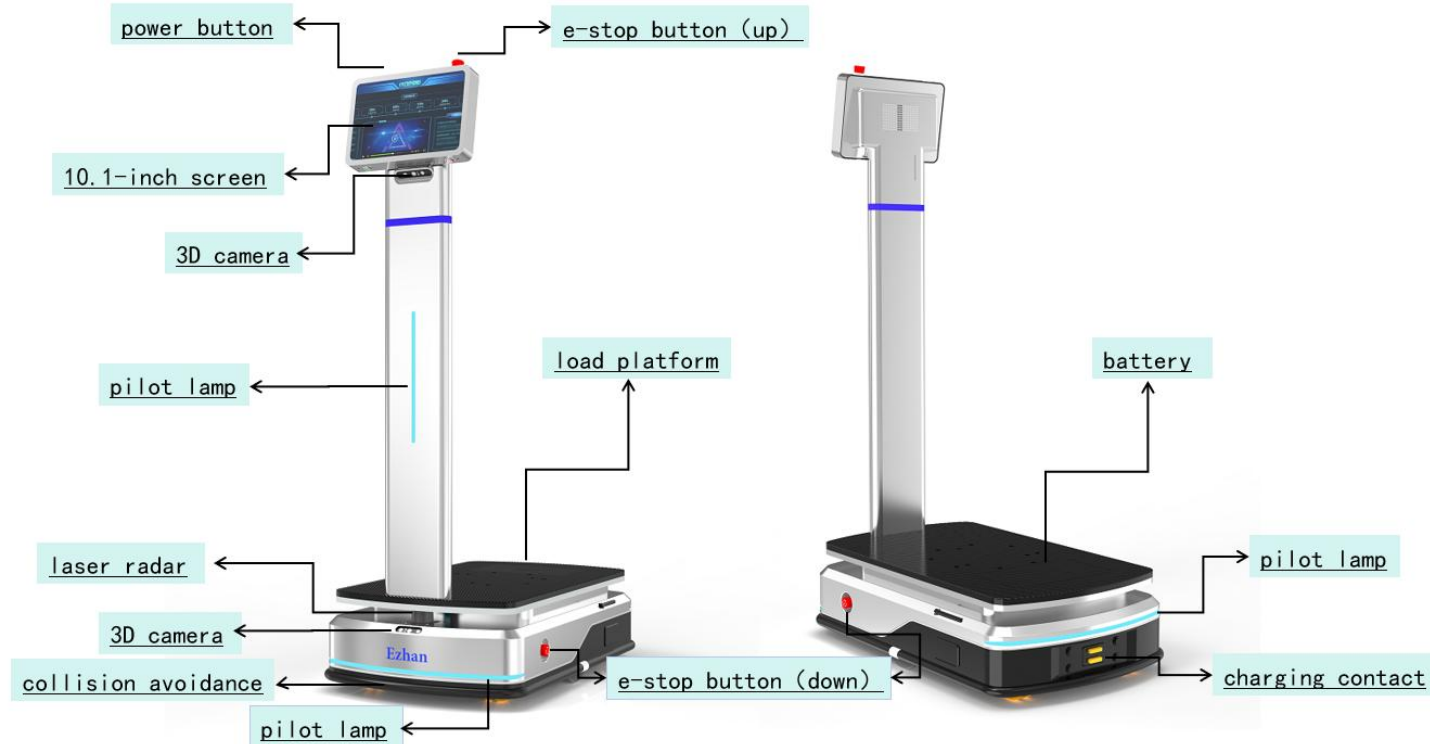
1. Selecting the Work Area

1-1. Unboxing



note

- ① Complete unboxing; ensure the AMR remains powered **OFF**.
- ② Familiarize yourself with the machine's component layout, as shown in the diagram on the left.

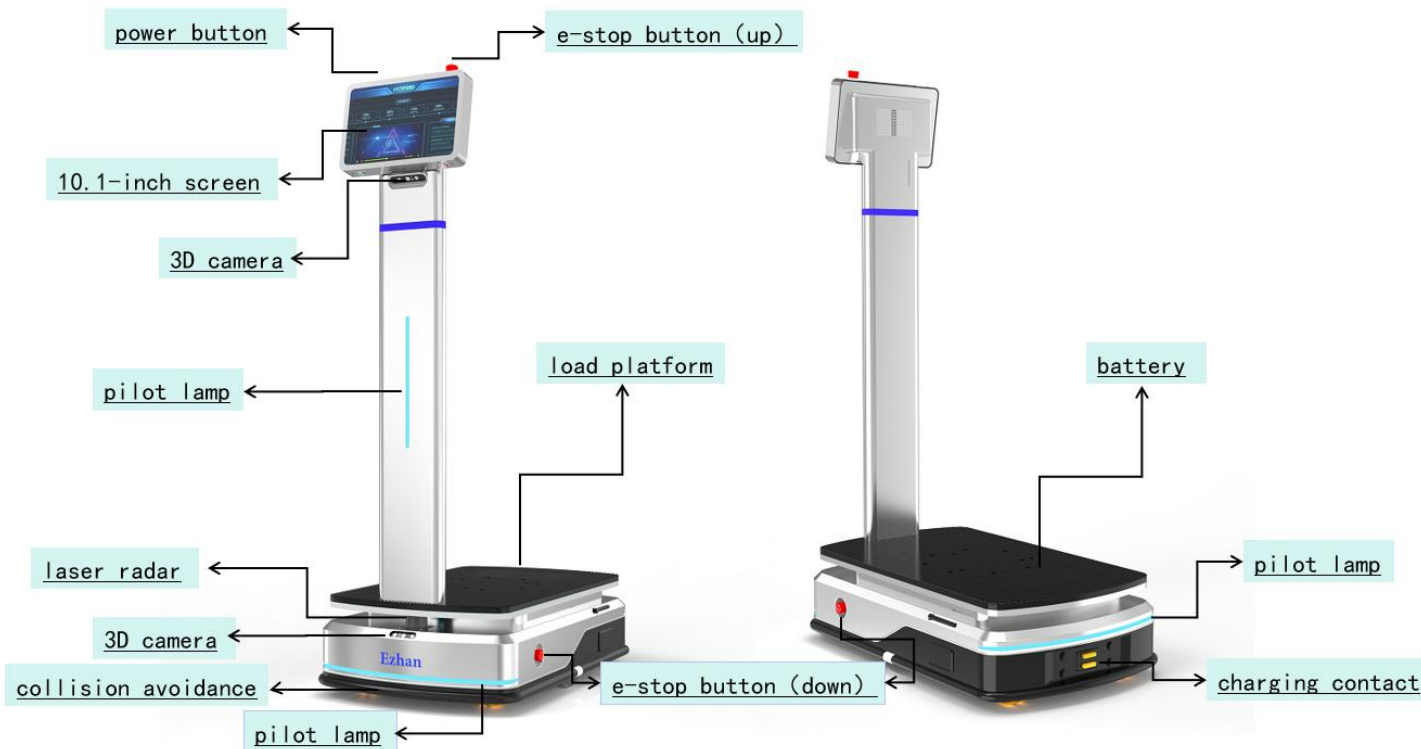


1. Selecting the Work Area

1-2. Selecting the Work Area



note



① Choose the operational work area for the AMR;

Tip: The selected area should have a flat, clean floor free of obstacles. Clear any debris that may interfere with the AMR's movement.

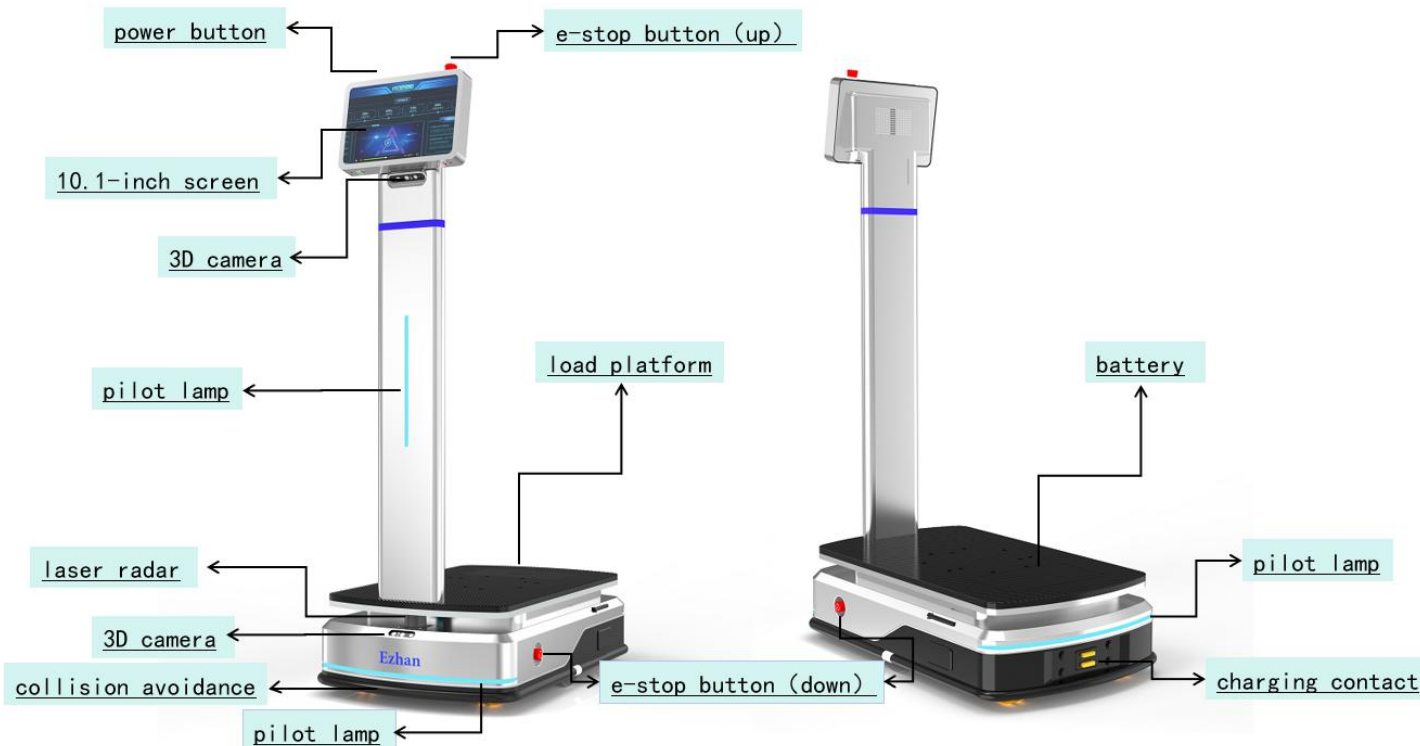


1. Selecting the Work Area

1-3. Moving the AMR to the Work Area



note



① While powered OFF, move the AMR to the designated work area.

Tip: When powered off, the AMR's drive wheels are unlocked and can move freely.



2. Power On

2-1. AMR Startup



note

- ① Press the Power Button.
- ② The Light Bar will illuminate.
- ③ The Screen will turn on.

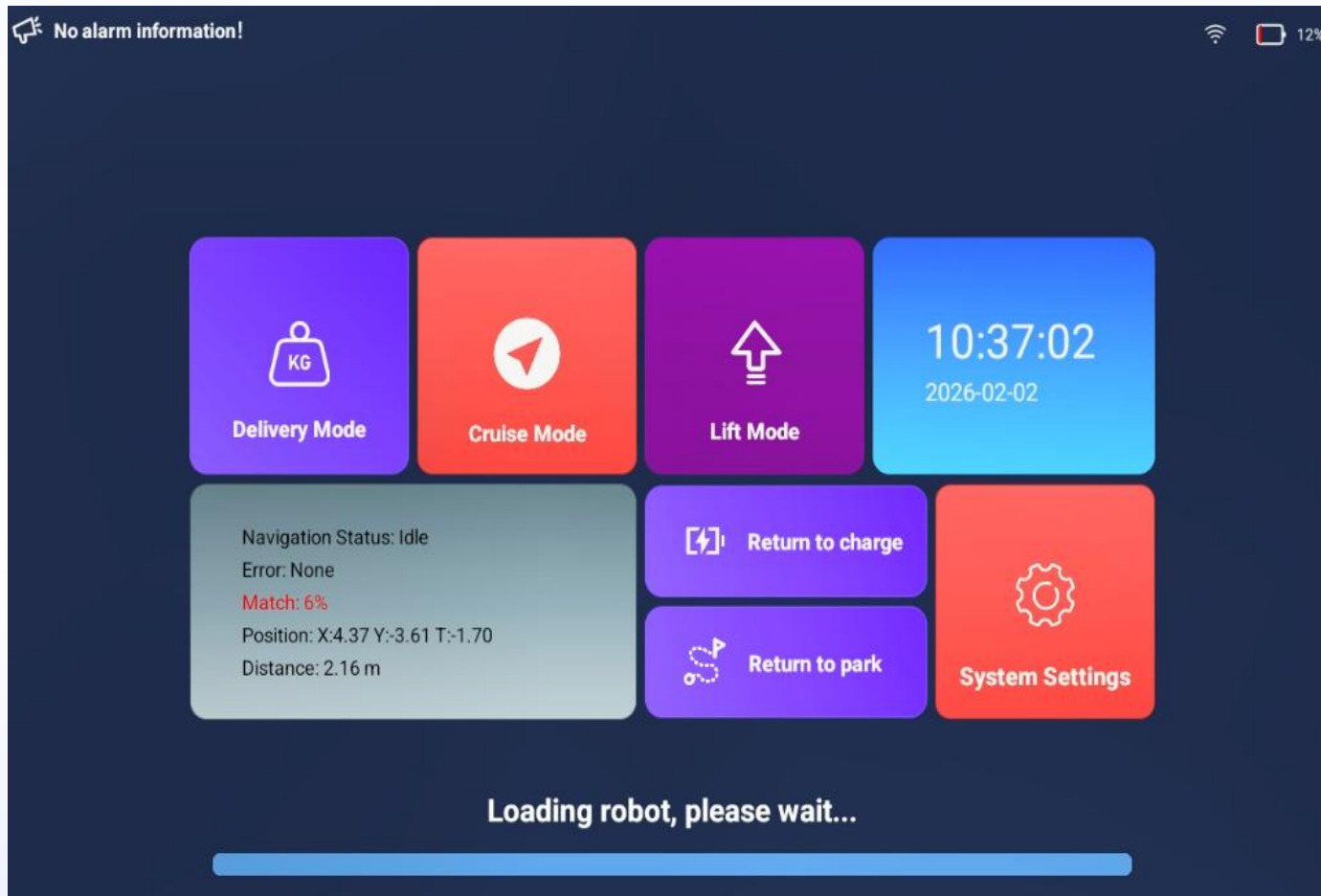


2. Power On

2-2. HMI Platform Startup



note



① Wait for the Main Interface to load completely.

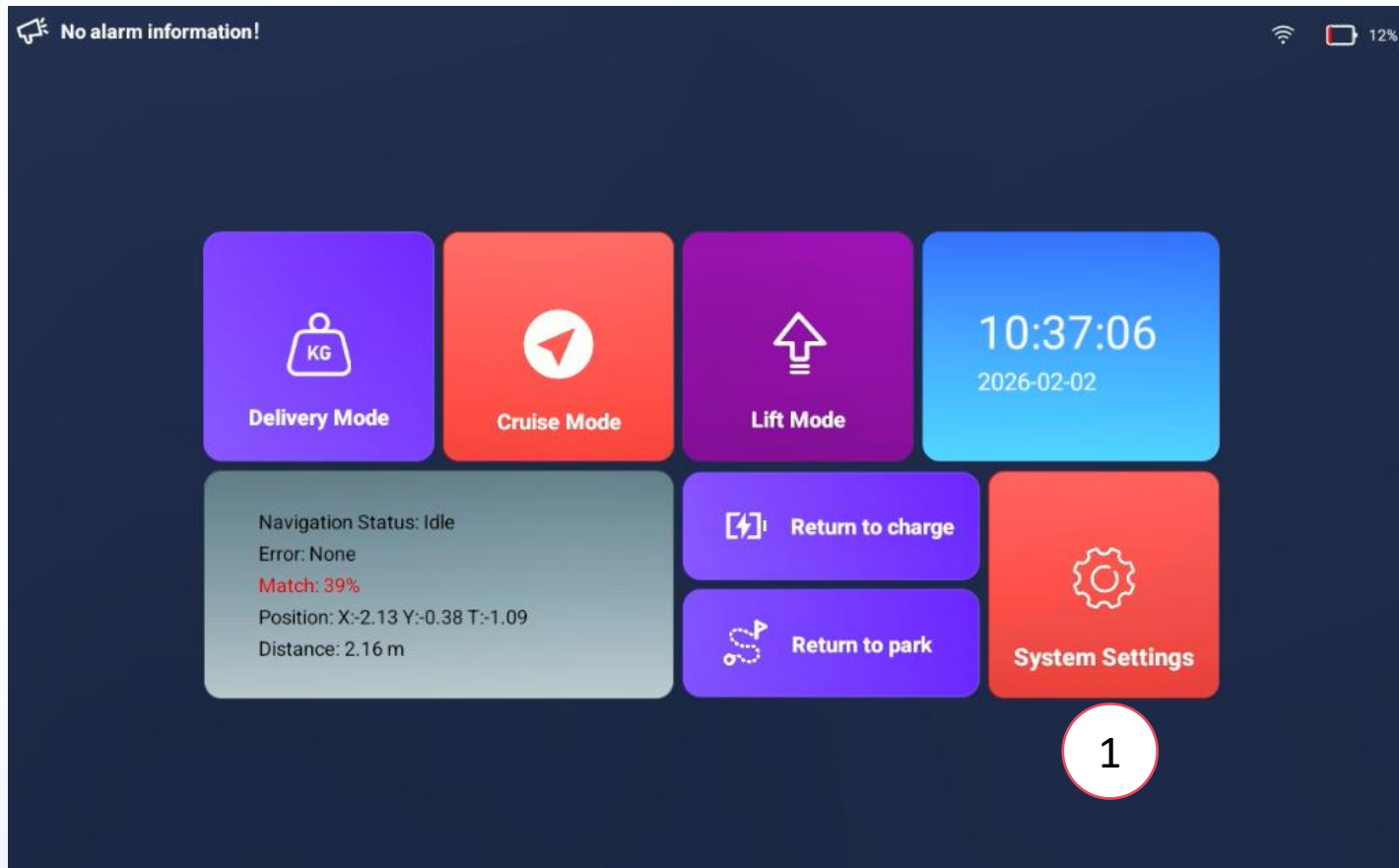
Tip: After startup, check if the AMR's light bar shows a steady red or if any exception voice alerts are played. If so, resolve the exception before proceeding.

Exceptions: Refer to the "Exception Handling" section at the end of this manual.



3. Mapping & Scanning

System Settings -Map Settings



note

- ① Access System Settings from the Main Interface.



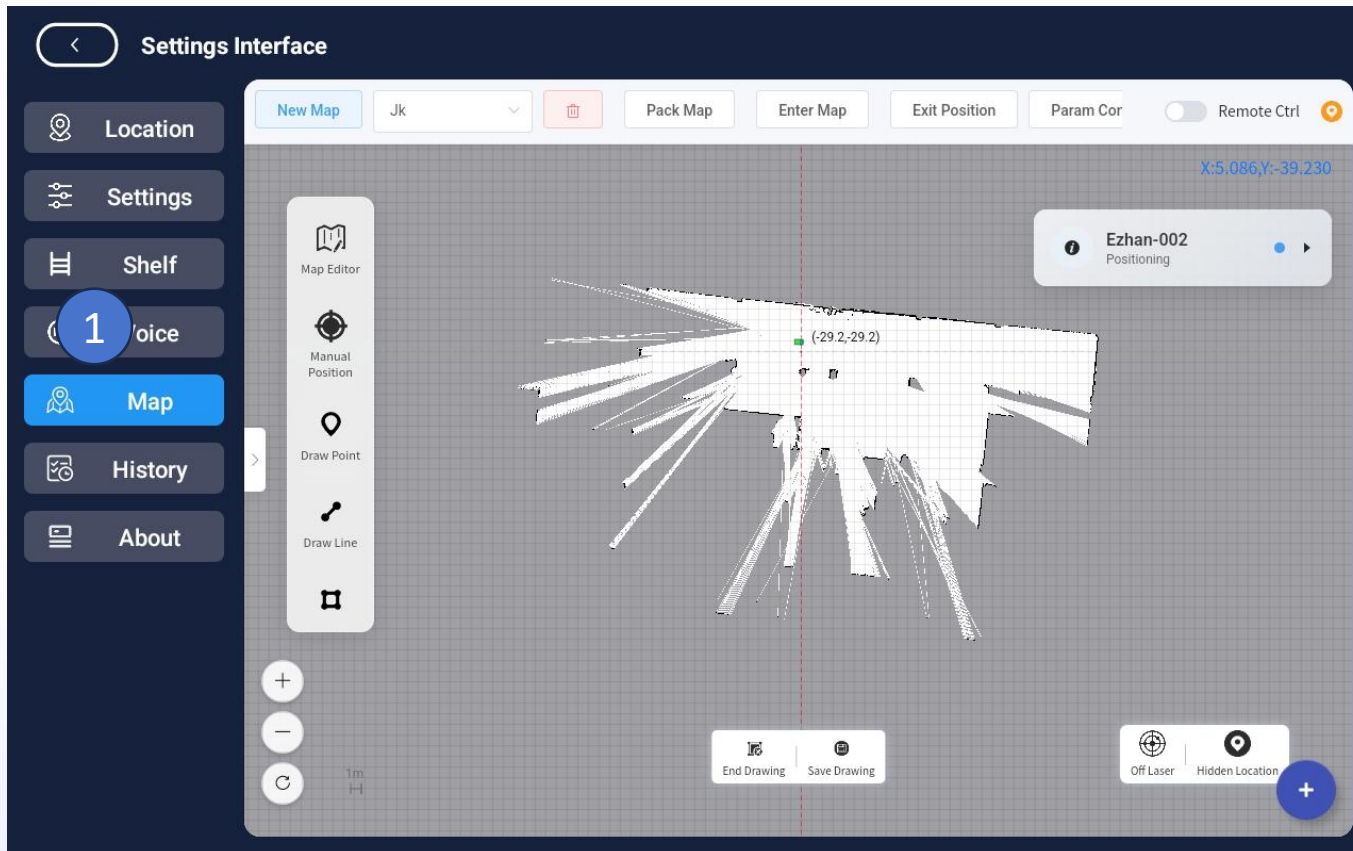
3. Mapping & Scanning

System Settings –Map Settings



note

① Navigate to Map Settings.

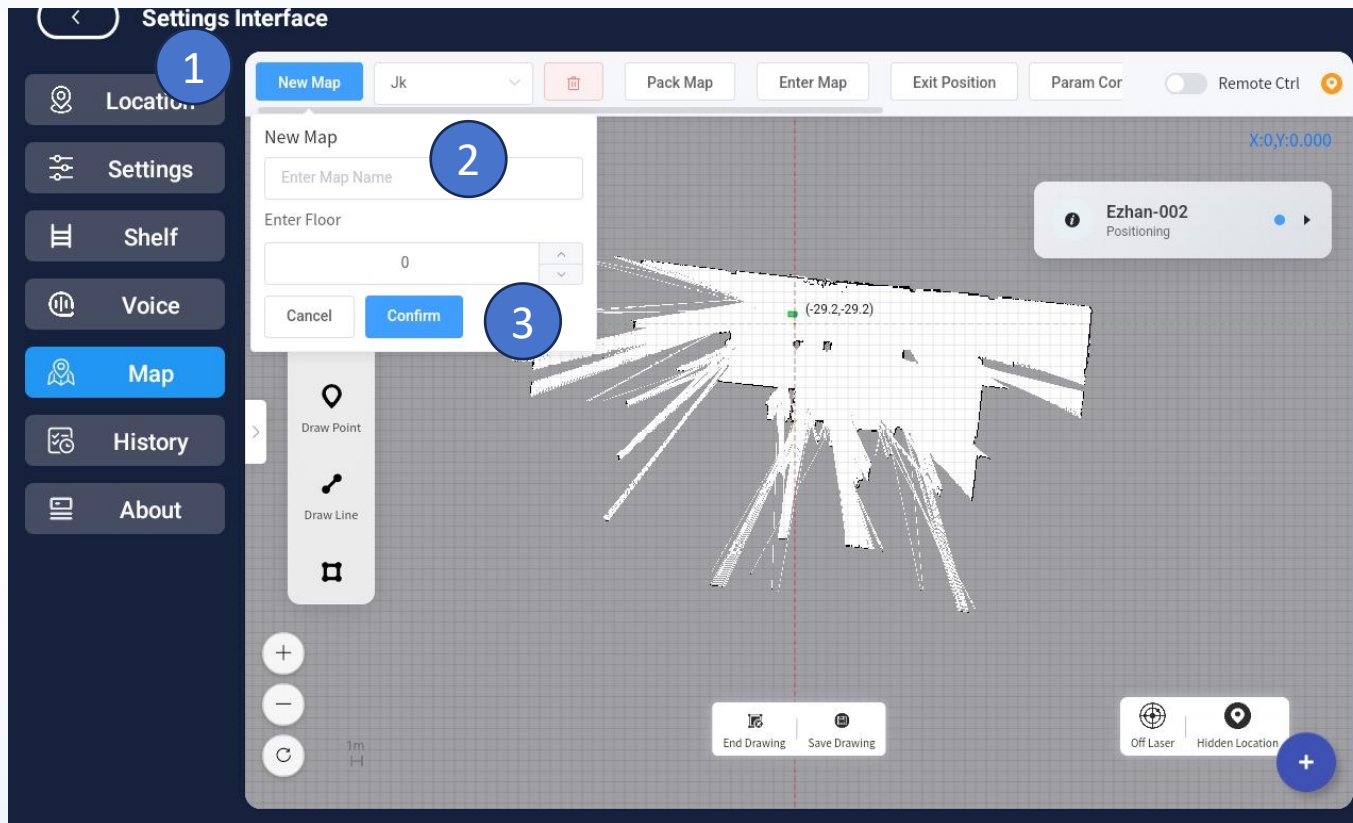


3. Mapping & Scanning

3-1. Create New Map



note



① Click " **New Map**".

② Enter Map Name.

③ Click "Confirm".

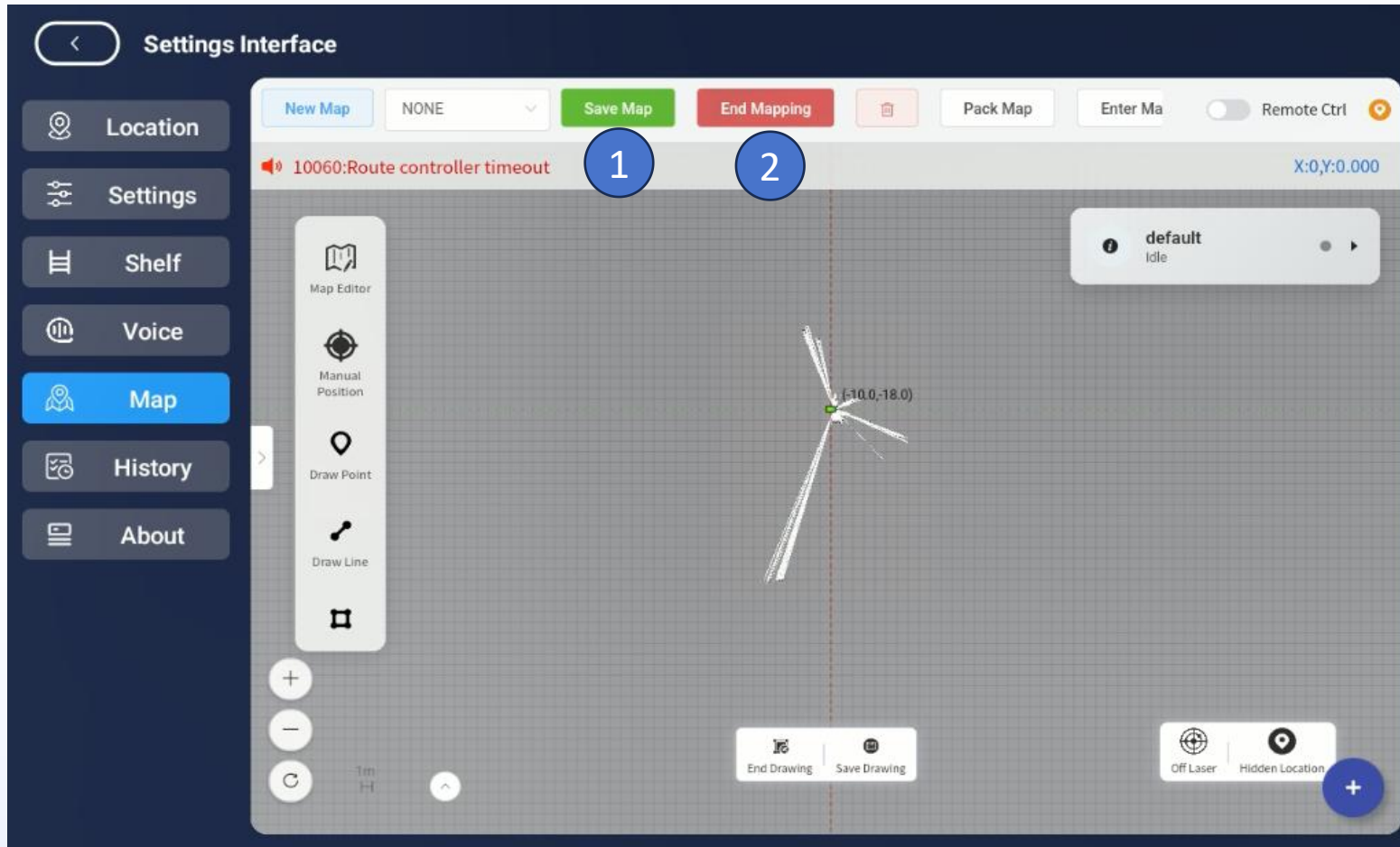


3. Mapping & Scanning

3-2. Scan Map



note



① With the **E-stop** pressed, push the AMR to map the area. Click "**Save Map**" when finished.

② Click "**End Mapping**" to exit the mapping interface.

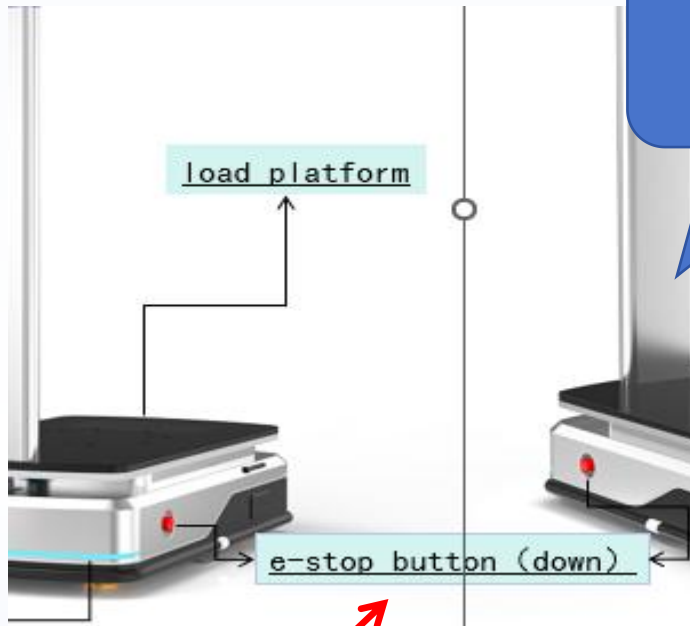


3. Mapping & Scanning



note

3-2. Scan Map



E-stop pressed;
drive wheels are
free to move.

Bottom Side E-Stop Button

- ① Powered ON: drive wheels locked, AMR is immobile.
- ② Side E-stop pressed: drive wheels unlocked.
- ③ With the E-stop activated, push the AMR to map. The red light bar and "E-stop triggered" voice prompt are normal.



3. Mapping & Scanning

3-2. Scan Map

E-stop not required when using the remote handle.



note

- ② Joystick
- ① Wake-up Button



- ① If a remote handle is available, it can also be used to control the AMR for mapping.
- ② Press wake-up button to pair. Flashing green light indicates success.
- ③ Steer the AMR using the joystick.

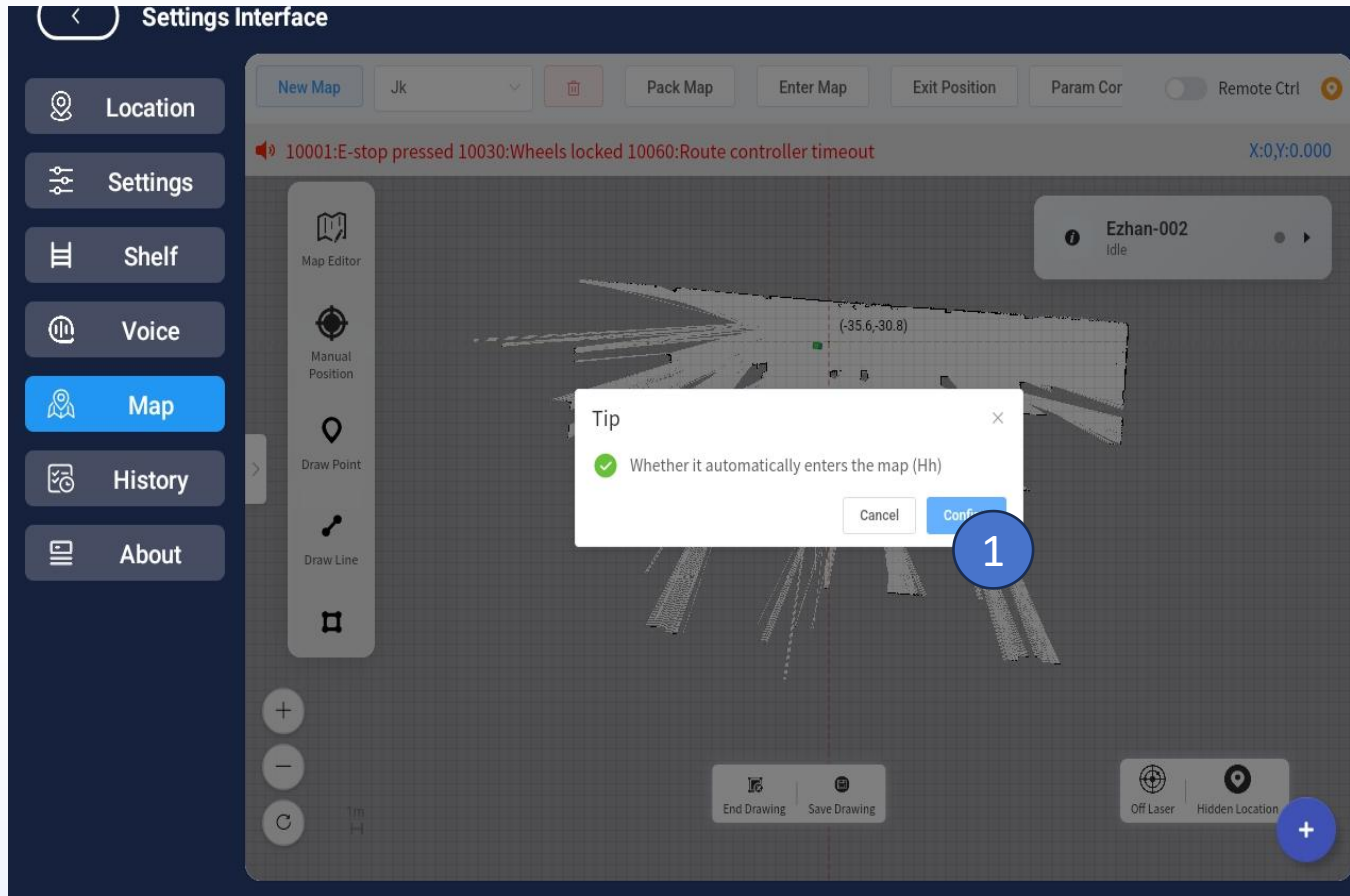


3. Mapping & Scanning

3-3. Enter Map



note



- ① Confirm the prompt after mapping to enter the map and initiate automatic relocalization.
- ② To manually access the map after cancellation, find it in the map list and select "Enter Map".



4. Setting Locations

4-1. Standard Locations



note



① Click “Work Locations”.

Process: Press the Side **E-Stop**, push the AMR to the desired location, and add waypoints.





4. Setting Locations



note

4-1. Standard Locations

Settings Interface

Work Locations

3 Add location

2 Location from map

5 Upload location

4

1	A	X: -1.079	Y: -0.346	Yaw: -0.136	6	Update
2	B	X: 1.373	Y: -0.619	Yaw: -0.055		Update
3	C	X: 3.252	Y: -0.718	Yaw: -0.054		Update
4	D	X: 4.291	Y: 0.463	Yaw: 1.554		Update
5	E	X: 2.558	Y: 0.994	Yaw: 3.135		Update

① Click "Work Locations".

② For new maps, click "Location from map" first to sync.

③ Move the AMR to the target spot on the map. Click "Add location", enter the location name.

④ Click "Update".

⑤ After adding locations, click "Upload location".

⑥ To delete a location, click the "Delete" icon.

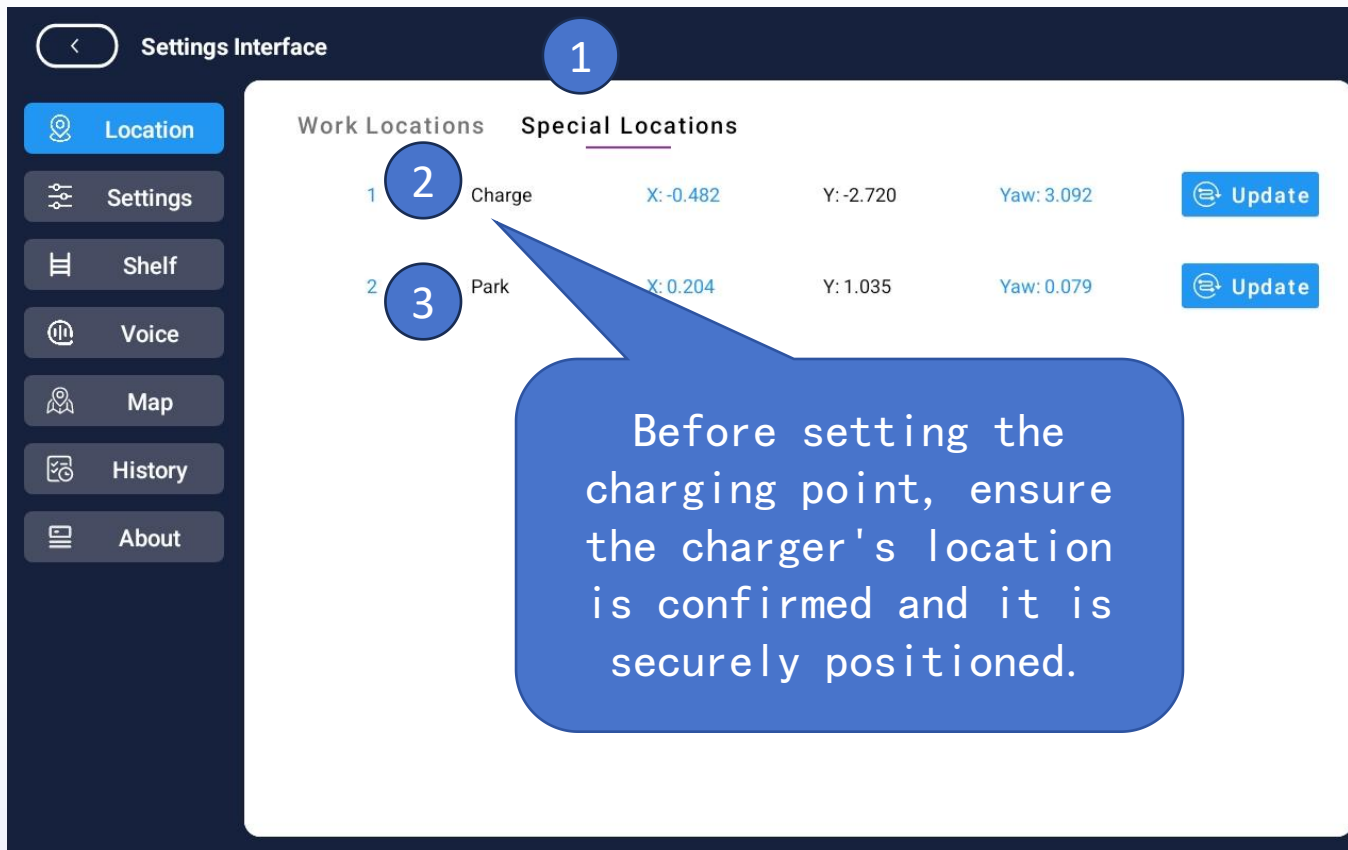


4. Setting Locations



note

4-2. Special Points



① Tap "**Special Locations**"

② For charging point setup: With E-stop pressed, align the charging port **2-5 cm** from the charger. Release E-stop, update location, then use Auto Charge from the main screen. Once charging, update the location again to complete.

③ To add a standby point: Press E-stop, move AMR to location, release E-stop, then update and confirm the location.

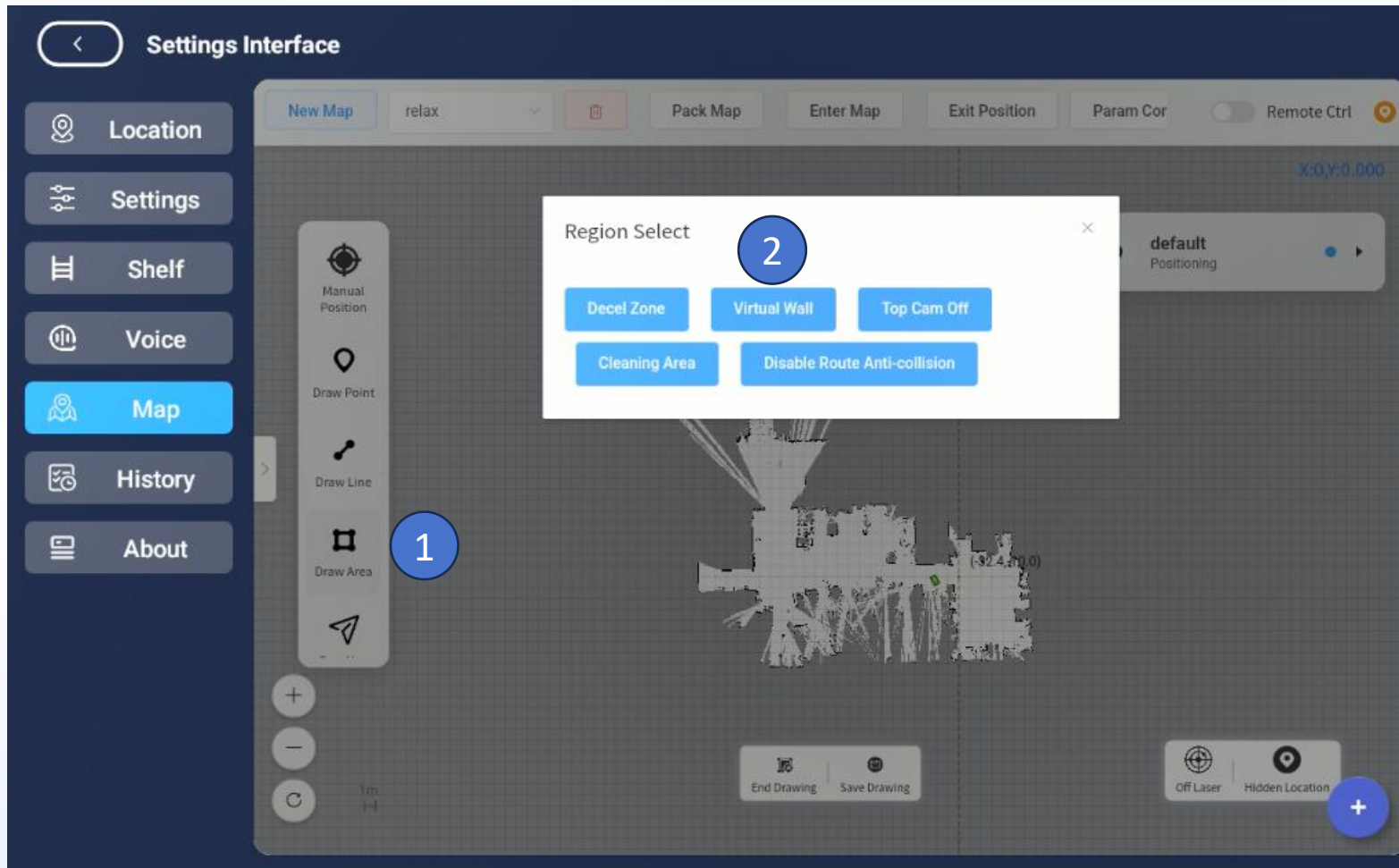


5. Setting Safety Zones

Safety Zone



note



- ① Click the "Draw Area" icon.
- ② Select "Virtual Wall" from the pop-up window.



5. Setting Safety Zones

Safety Zone



note

Safety Zones: Define areas on the map where the AMR is prohibited from traveling.

Zoom the map to a suitable size (pinch to adjust), then draw the safety zone.



- ① Place points on the map to define the 4 vertices of the safety area (**clockwise or counterclockwise**).
- ② Click "**Save Drawing**".
- ③ Click "**End Drawing**" to exit.

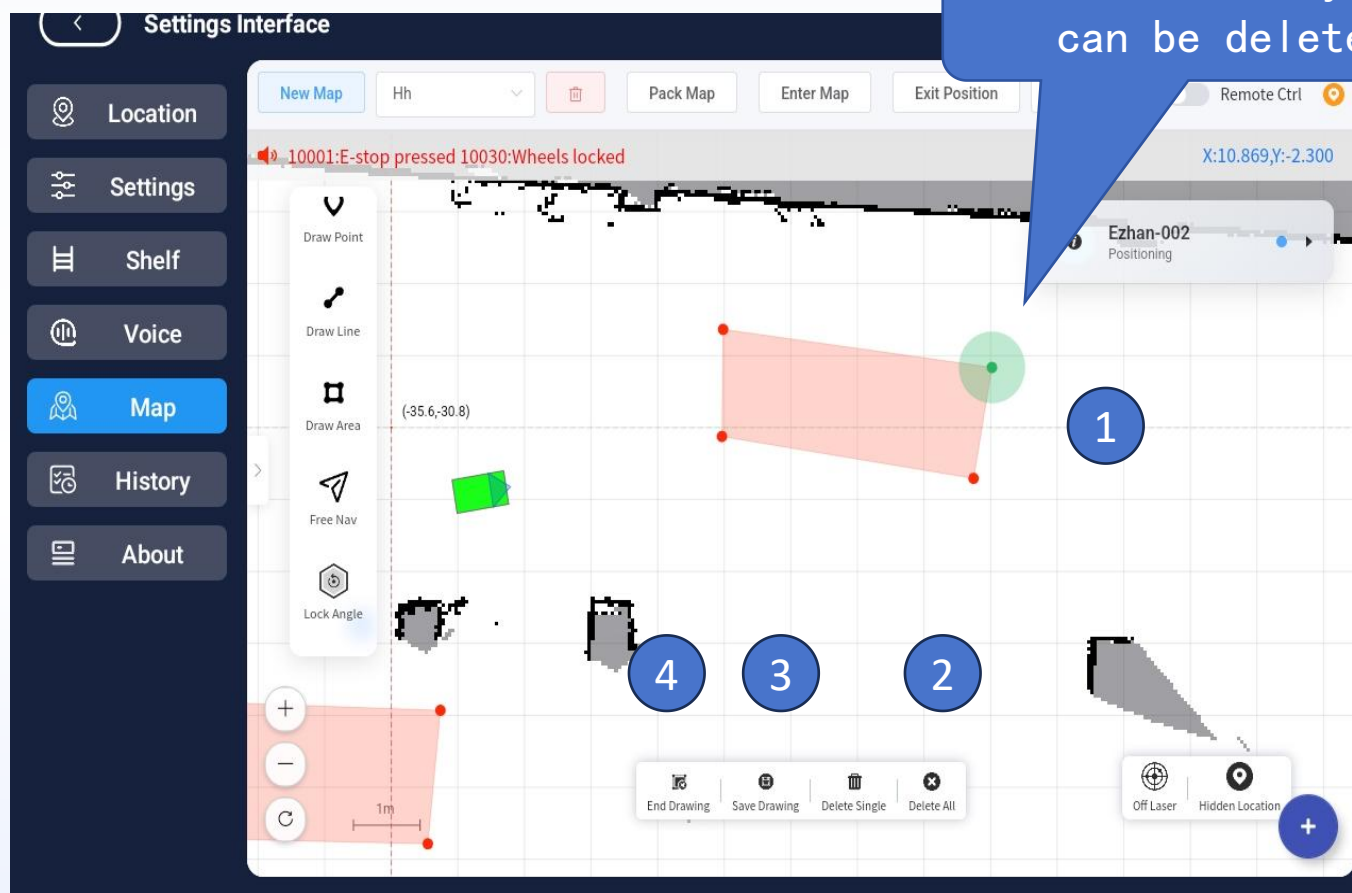


5. Setting Safety Zones

Safety Zone



note



- ① Click to select one vertex of the safety zone.
- ② Click "**Delete All**";
- ③ Click "**Save Drawing**";
- ④ Click "**End Drawing**" to exit.



6. Selecting Task Mode

Select the corresponding task mode



A –Delivery Mode: For moving materials placed on the AMR's cargo platform. Enables single or multi-point material transport.



B –Cruise Mode: For carrying materials, enables multi-point regular or timed cyclic task execution.



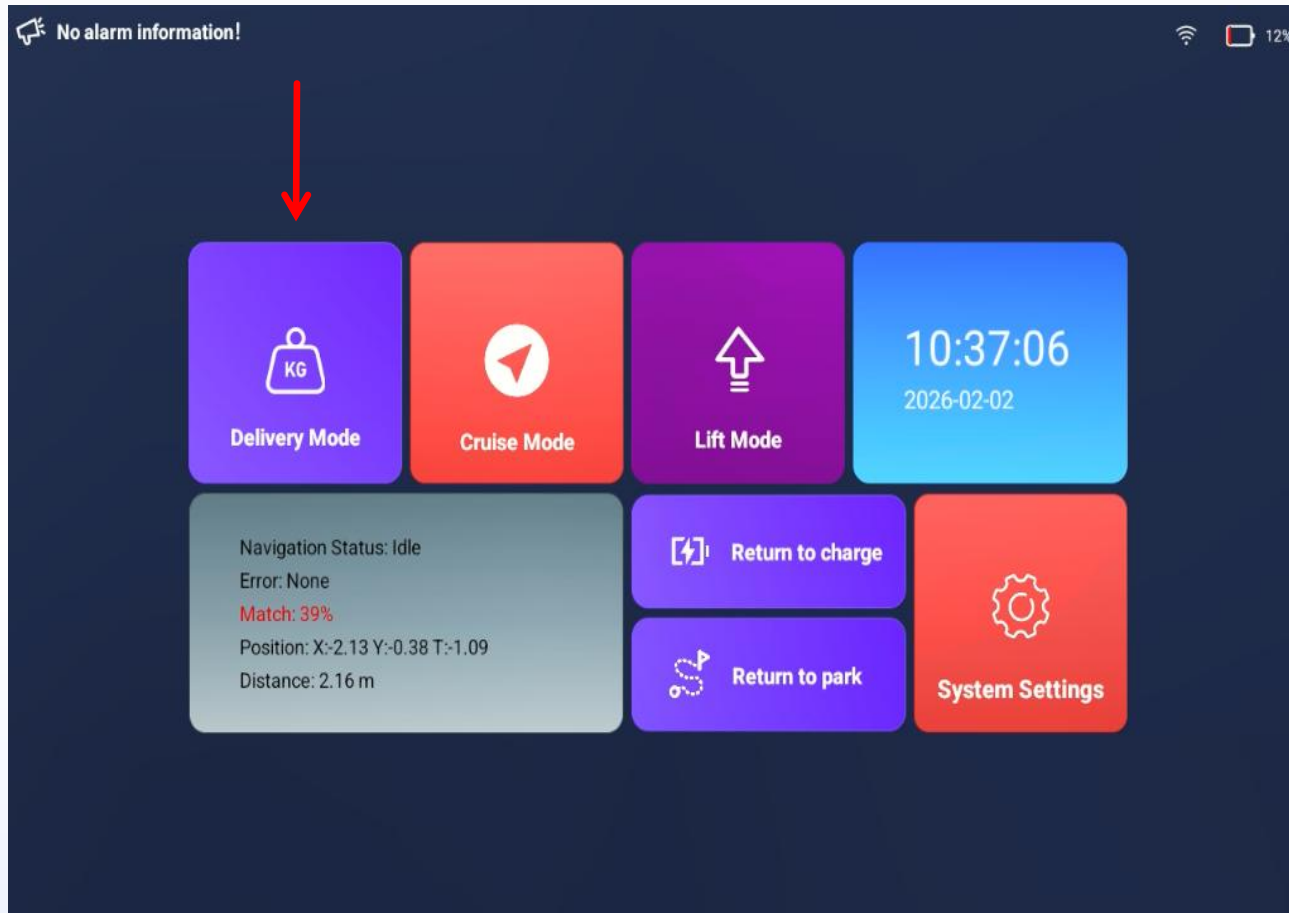
C –Lift Mode: For AMR identifying and lifting/placing racks. Enables point-to-point rack transport.

6. Selecting Task Mode

A -Delivery Mode



note



Operation: Select 'Delivery Mode' on the main screen. The AMR will auto-navigate to chosen points to execute transport tasks.

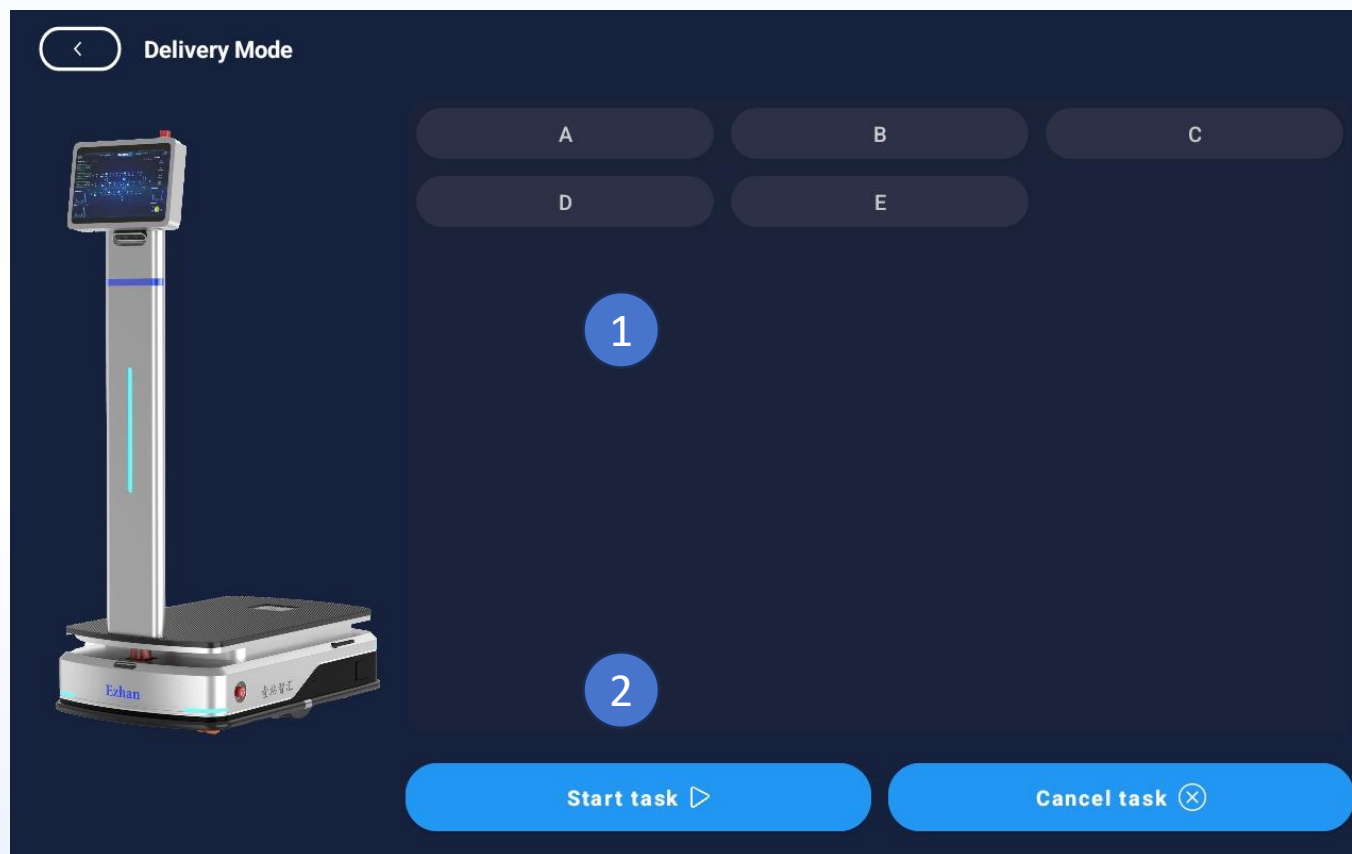


6. Selecting Task Mode

➤➤➤ A -Delivery Mode



note



- ① Select one or multiple designated points.
- ② Click "**Start Task**".



6. Selecting Task Mode

➤ A -Delivery Mode



note



Navigating to task location: A

- ① The AMR begins task execution.
- ② During Task: Lightly touch the screen to pause.



6. Selecting Task Mode

➤➤➤ A -Delivery Mode



note

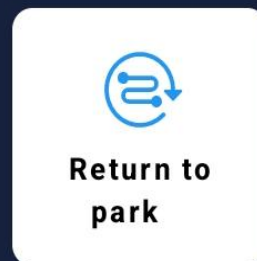
Task Paused



1



2



3

- ① Click "End" to cancel task.
- ② Click "Resume" to resume.
- ③ Click "Return to park" to park.

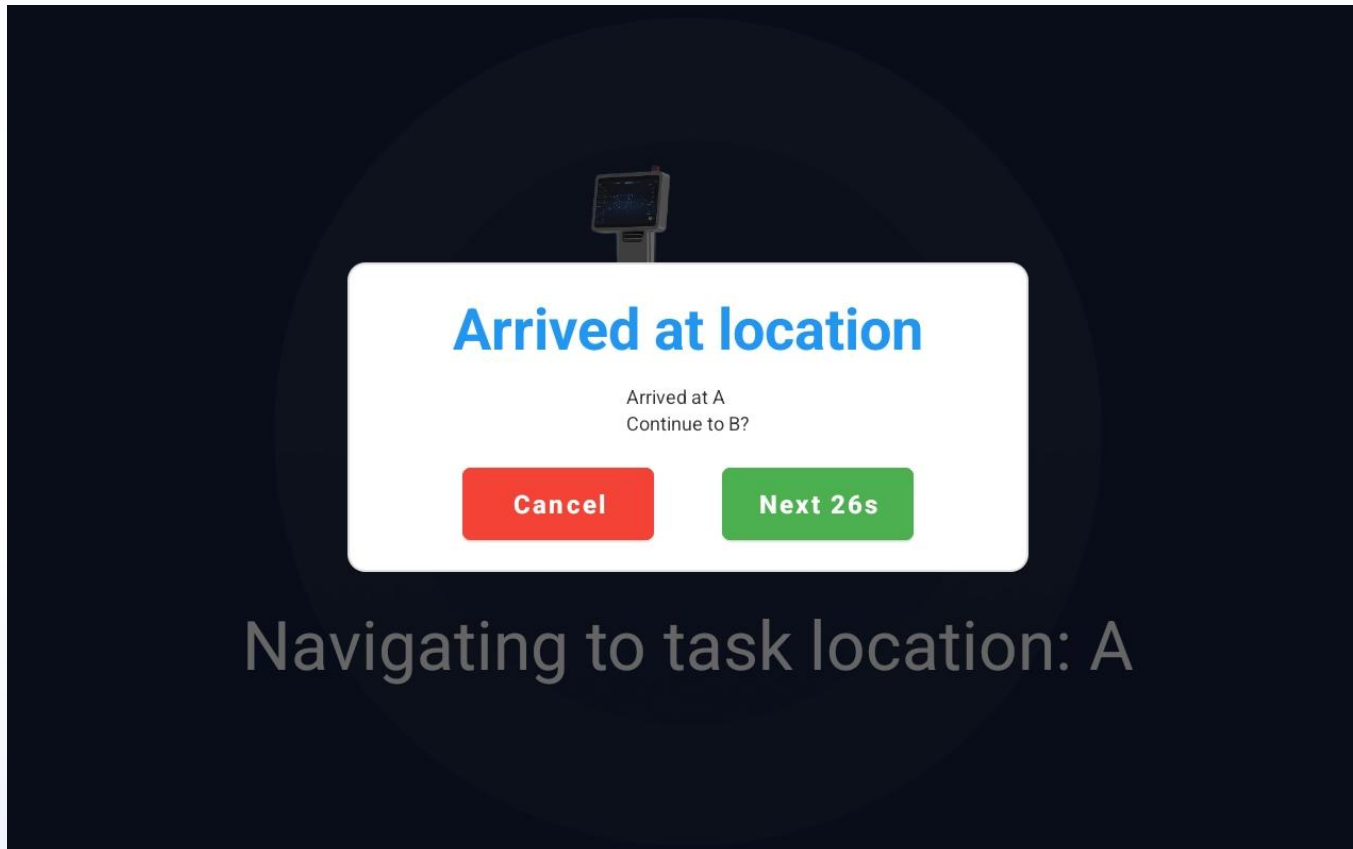


6. Selecting Task Mode

➤ A -Delivery Mode



note

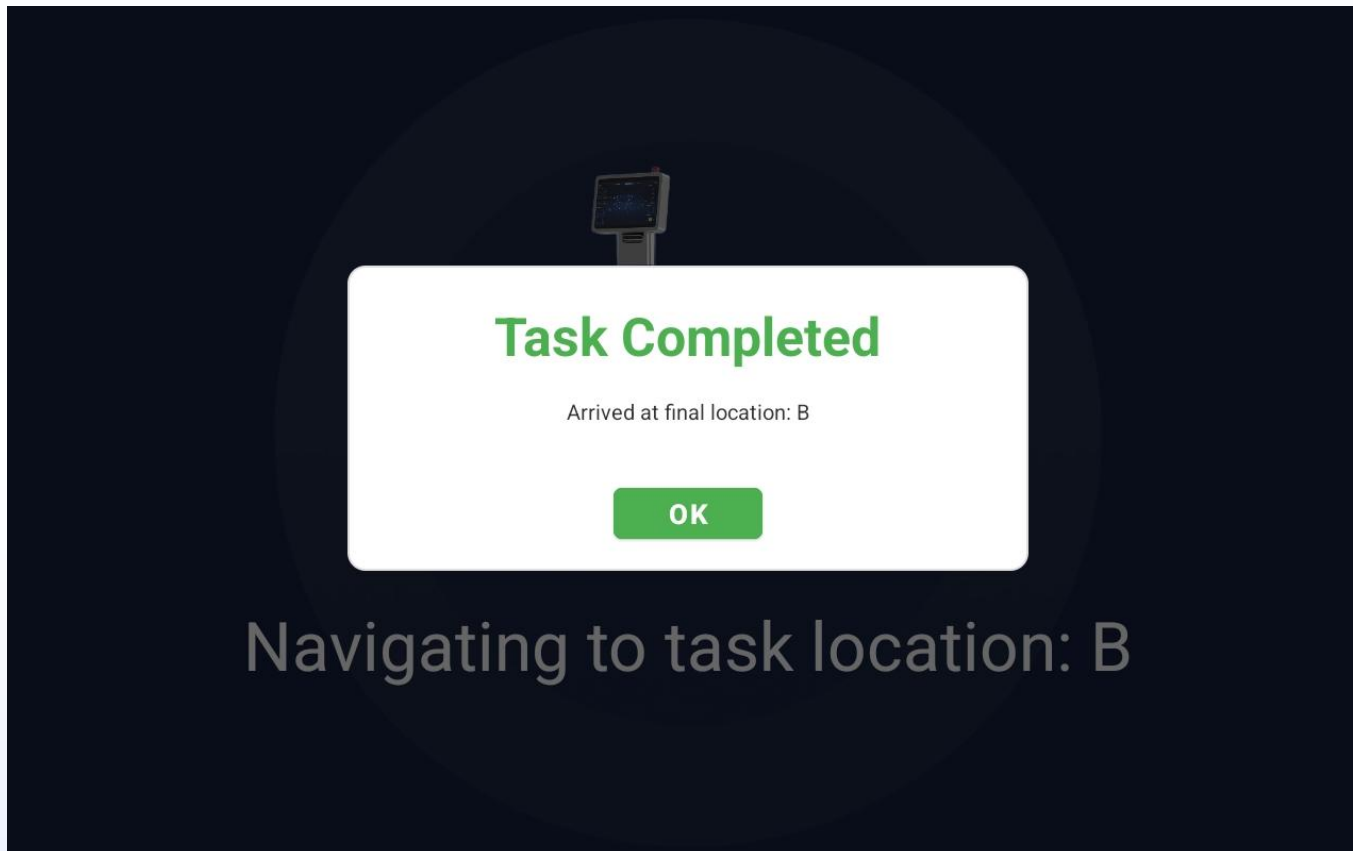


- ① At each point, AMR waits 60 seconds (default) before proceeding to the next.
- ② For multiple points, AMR follows the selected order.



6. Selecting Task Mode

➤ A -Delivery Mode



- ① Upon reaching the final point, "Task Completed" is prompted. Click "OK".

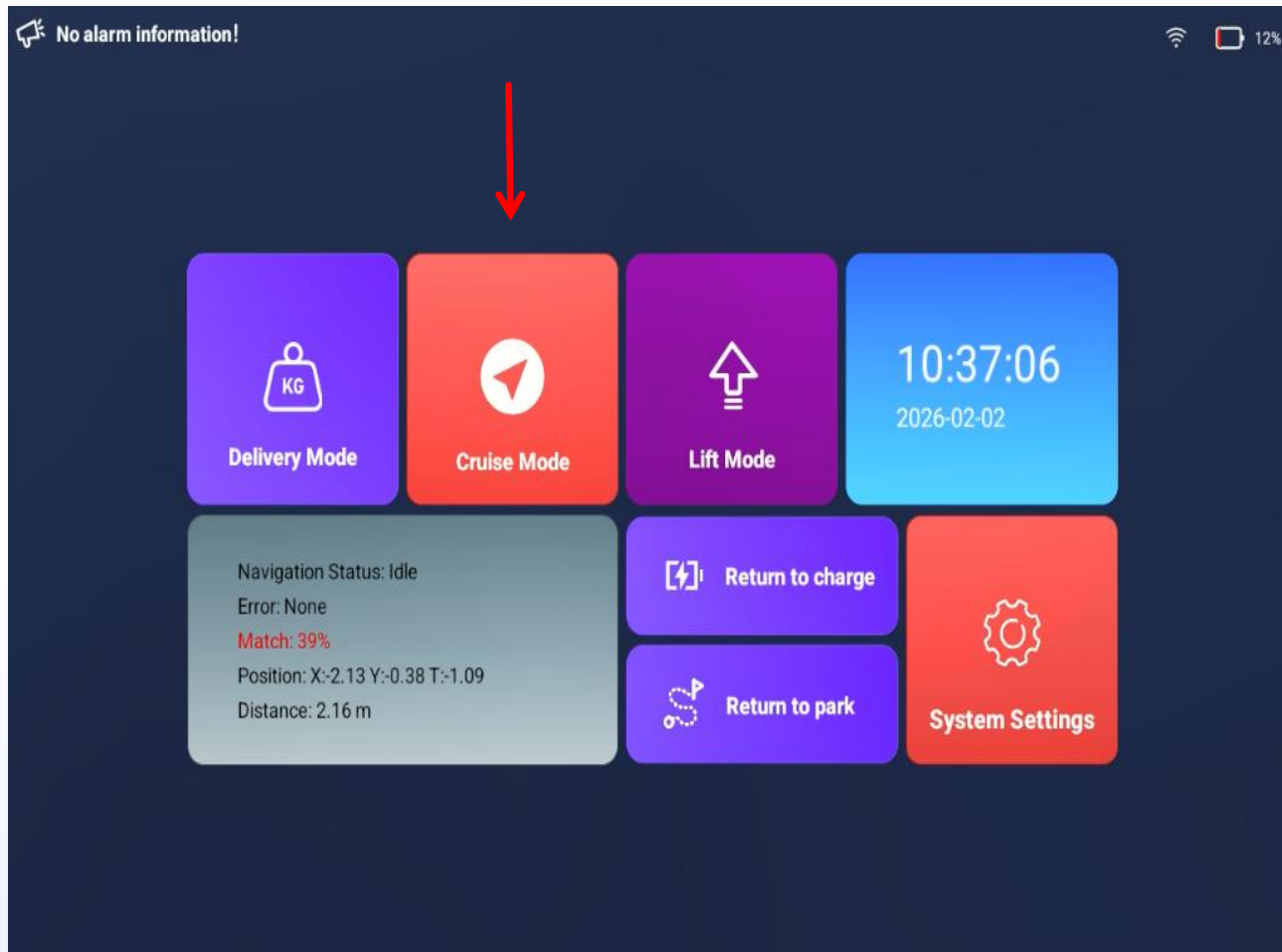


6. Selecting Task Mode

B –Cruise Mode



note



Operation Overview: On the main interface, tap “Cruise Mode” to enter its interface. Using default settings, the AMR will automatically route to the selected locations and perform cyclic operations between multiple points.

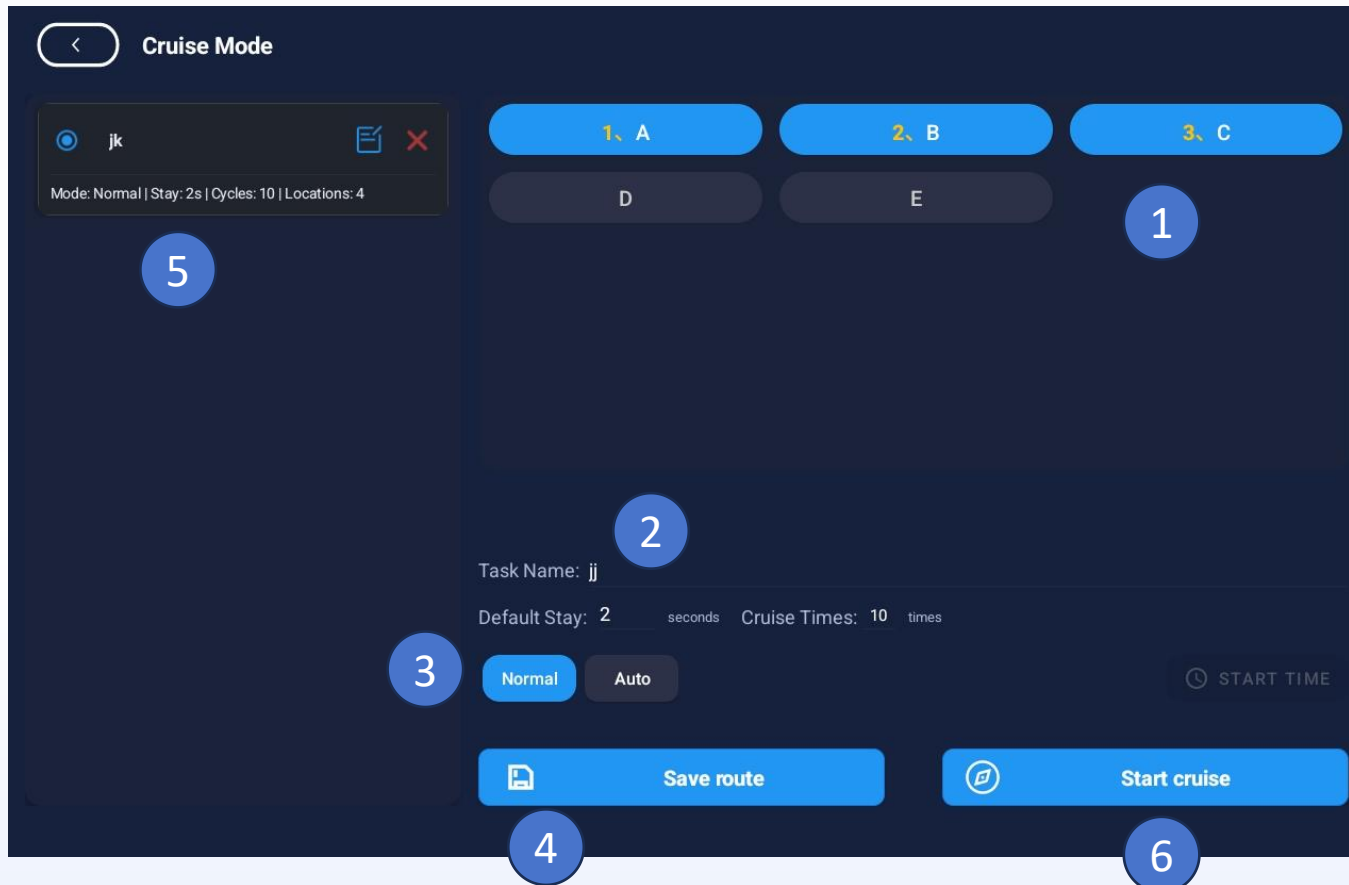


6. Selecting Task Mode



note

B –Cruise Mode –Regular Patrol



- ① Select points.
- ② Enter Task Name, set Wait Time at each stop, set Repetitions.
- ③ Click “Normal” mode.
- ④ Click "Save route". Saved tasks appear in the left panel.
- ⑤ **Select a saved task.**
- ⑥ Tap "Start cruise" to begin the task, during which the AMR will navigate to each selected location, repeat the set number of cycles, and wait for the specified seconds at each stop.

6. Selecting Task Mode

➤➤➤ B -Cruise Mode -Regular Patrol



note



Navigating to task location: A

- ① The AMR begins the patrol task.
- ② During task execution, tap anywhere on the screen to pause the task.

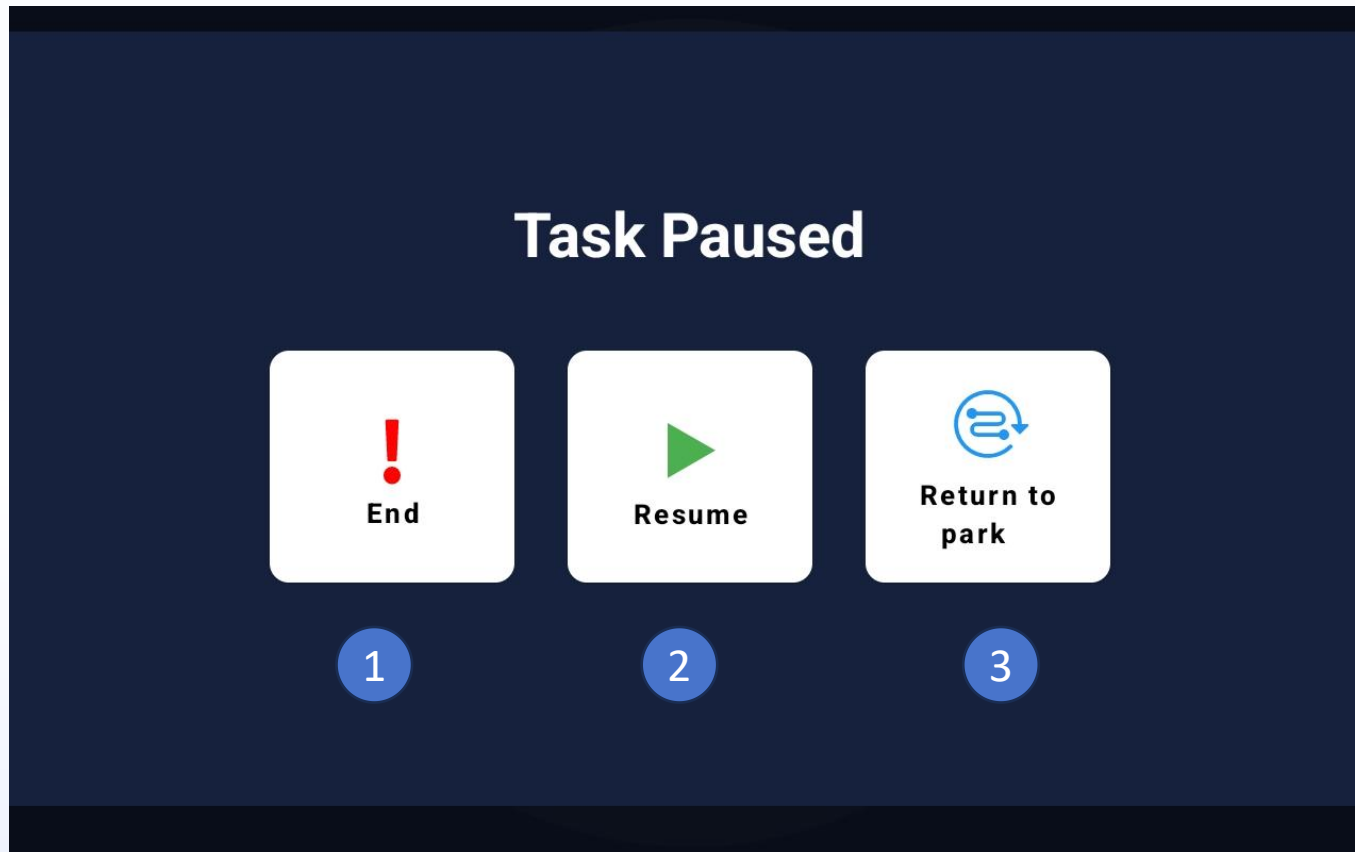


6. Selecting Task Mode

➤➤➤ B -Cruise Mode -Regular Patrol



note



① Tap "End" to cancel the task

② Tap "Resume" to resume.

③ Click "Return to park" to park.

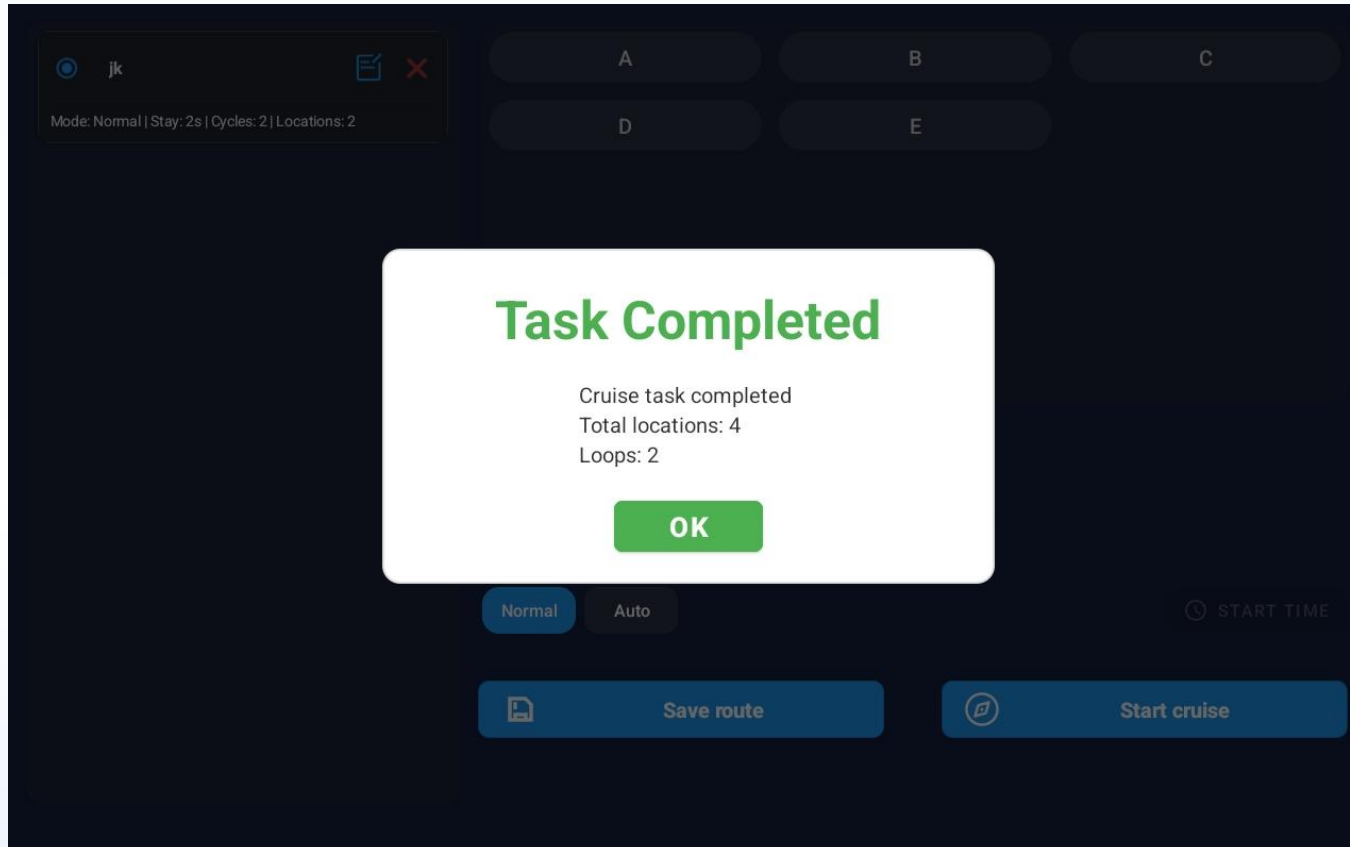


6. Selecting Task Mode

➤➤➤ B -Cruise Mode -Regular Patrol



note



When the AMR reaches the final location, a task completion prompt will appear. The user can tap "OK" to confirm completion.

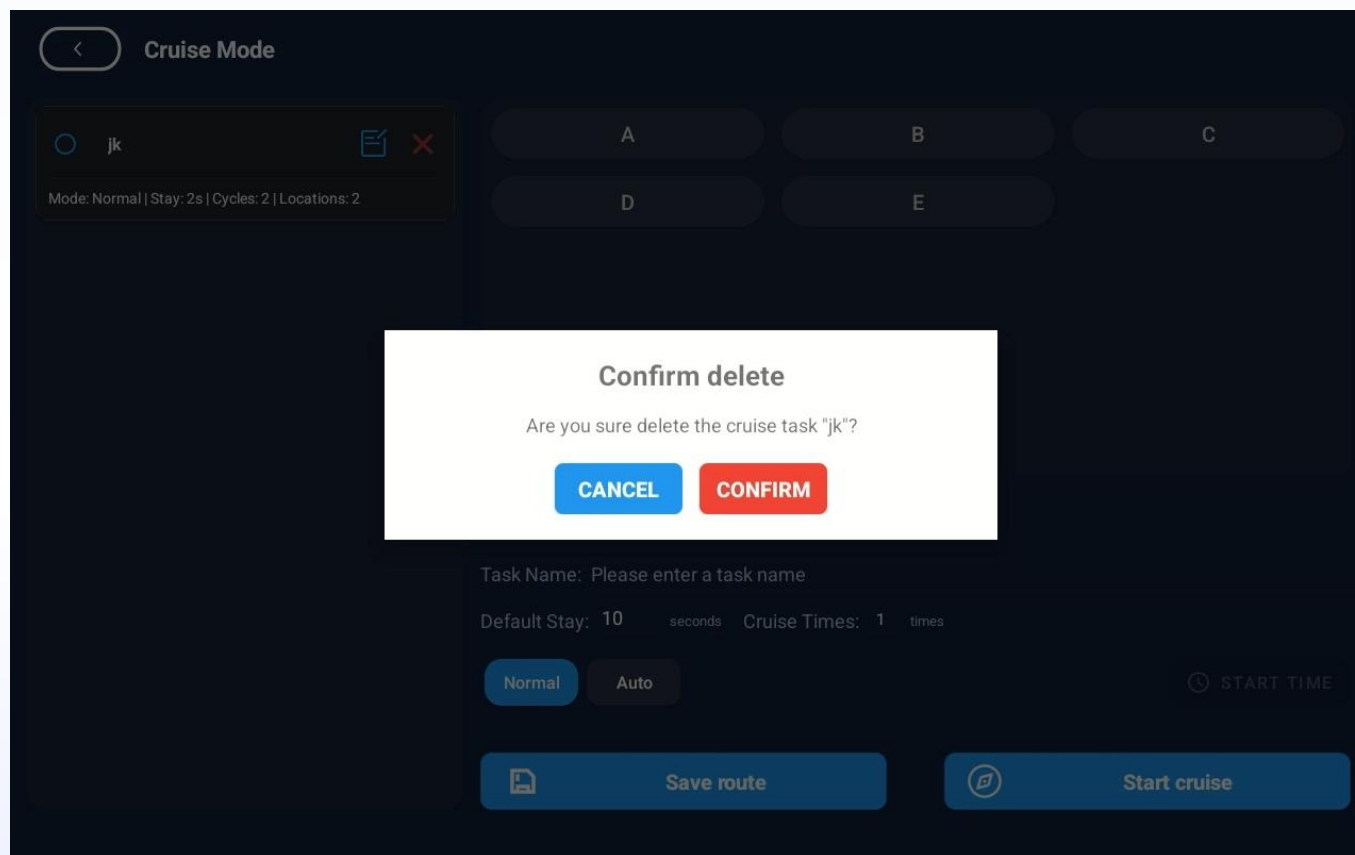


6. Selecting Task Mode

➡➡➡ B -Cruise Mode -Standard Patrol (Delete)



note



To delete a patrol task, click the "X" in the left panel and confirm the action in the pop-up window.



6. Selecting Task Mode

B -Cruise Mode -Standard Patrol (Edit)



note

The screenshot shows the 'Cruise Mode' interface. At the top left, there is a back arrow and the text 'Cruise Mode'. Below this, a task card for 'jk' is shown with details: 'Mode: Normal | Stay: 2s | Cycles: 2 | Locations: 2'. A blue circle with the number '1' points to the task card. To the right of the task card, there are five location buttons: '1. A', '2. B', 'C', 'D', and 'E'. A blue circle with the number '2' points to the '2. B' button. Below the location buttons, there is a section for task configuration. A blue circle with the number '3' points to the 'Task Name: jk' field. Below this, there are fields for 'Default Stay: 2 seconds' and 'Cruise Times: 2 times'. There are two buttons, 'Normal' and 'Auto', with 'Normal' selected. A 'START TIME' button is also present. At the bottom, there are two large blue buttons: 'Update route' and 'Start cruise'. A blue circle with the number '4' points to the 'Update route' button.

- ① Click the "Edit" icon for the desired task.
- ② Reselect or modify one or more designated locations.
- ③ Modify the task name, set the wait time for each pause, and configure the number of repetitions.
- ④ Click "Update route" to save the changes. The updated task will then appear in the left panel.

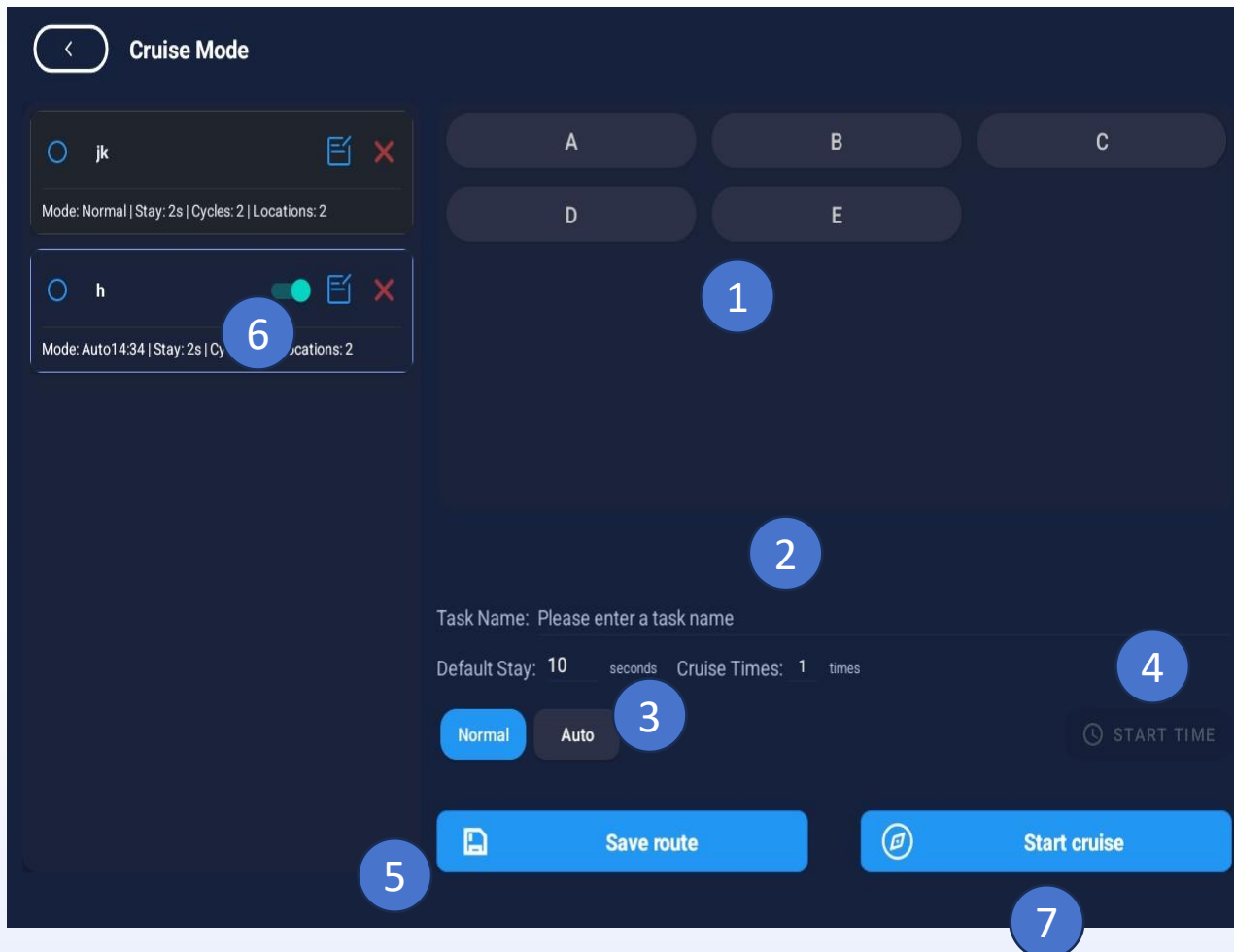


6. Selecting Task Mode



note

B -Cruise Mode -Automatic Patrol



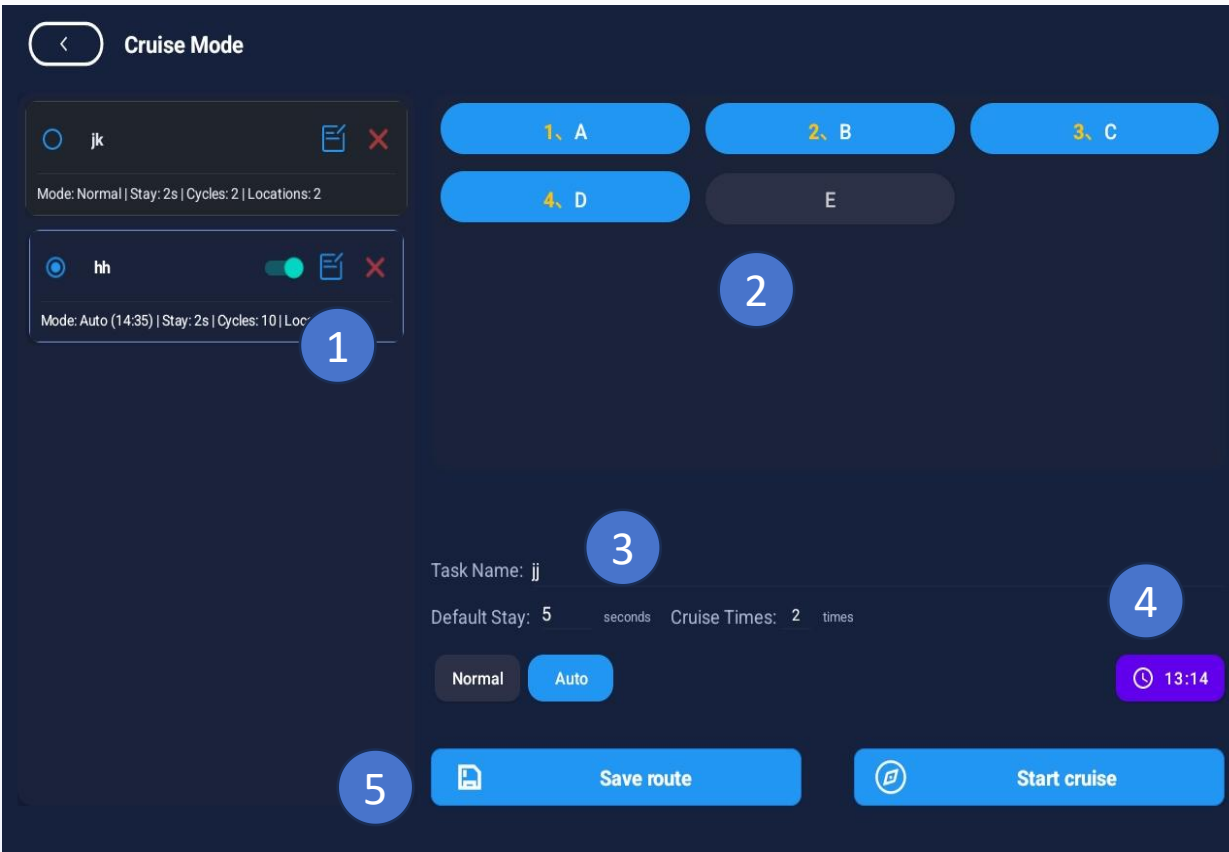
- ① Select one or more designated locations.
- ② Enter a task name, set the wait time for each pause, and configure the number of repetitions.
- ③ Click "Auto" to enter automatic mode.
- ④ Click "START TIME" to set the task's scheduled start time.
- ⑤ Click "Save route" to store the patrol task. The saved task will appear in the left panel.
- ⑥ Open the saved Automatic Patrol task.
- ⑦ Click "Start cruise" to begin. The AMR will commence at the scheduled time, navigate to all selected locations, repeat the set number of cycles, and wait the specified seconds between repetitions.

6. Selecting Task Mode



note

B -Cruise Mode -Automatic Patrol (Edit)



- ① Click the "Edit" icon for the target task.
- ② Reselect or modify one or more designated locations.
- ③ Update the task name, set the wait time for each pause, and adjust the number of repetitions.
- ④ Modify the task's scheduled start time.
- ⑤ Click "Update route" to save changes. The updated task will appear in the left panel.

6. Selecting Task Mode

B –Patrol Mode –Automatic Cruise

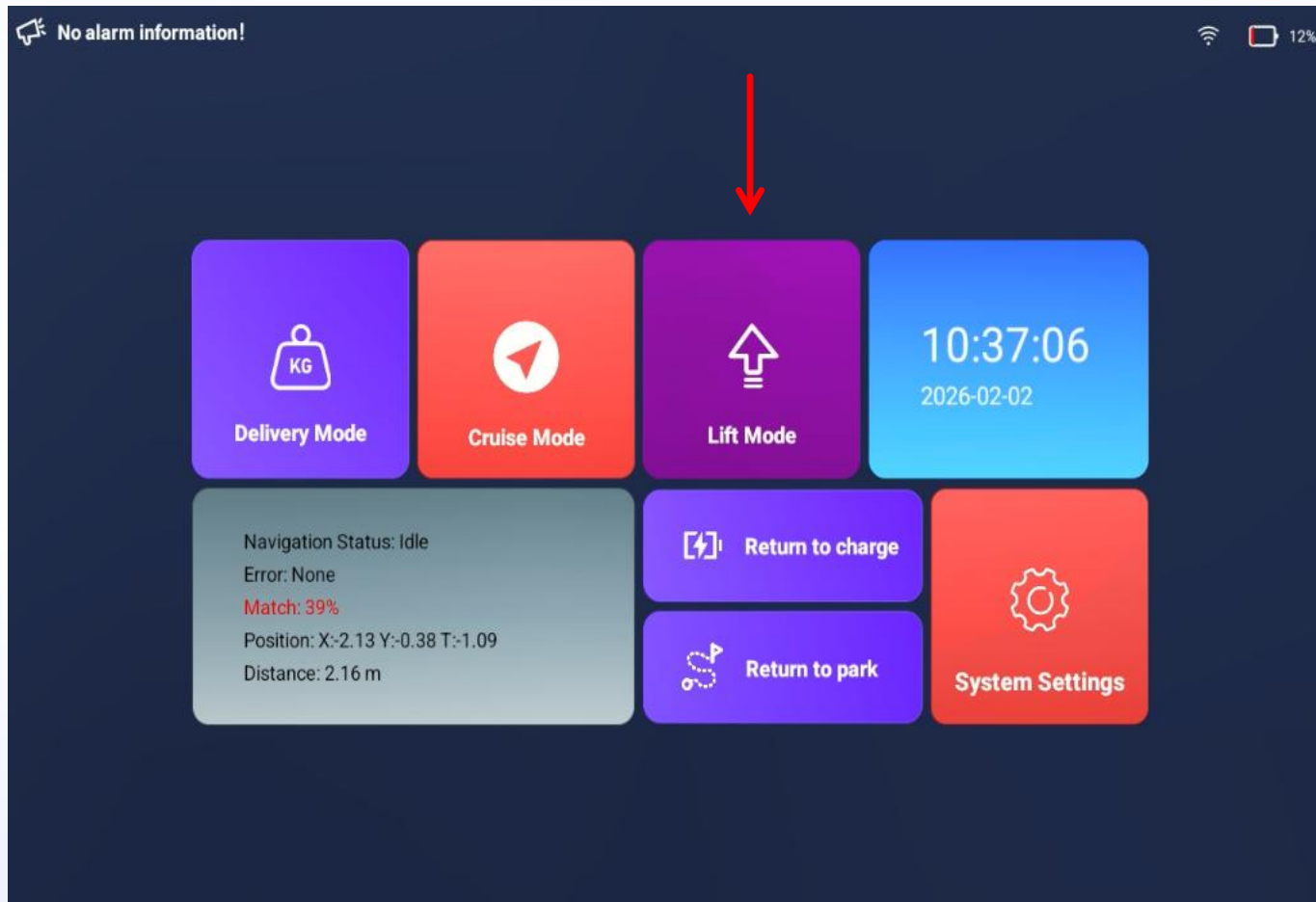
- The task will start automatically when the scheduled time is reached.
 - During task execution, tap anywhere on the touchscreen to pause the task. Tap "End" to cancel the task or "Continue" to resume execution.
 - When the AMR reaches the final location, a task completion prompt will appear. The user can tap "OK" to confirm that the task is complete.
- Note 1: If a task is sent by the host before the scheduled start time, the AMR will execute that task first.
 - Note 2: If other tasks are still in progress when the scheduled start time is reached, the AMR will complete them before initiating the scheduled patrol task.
 - Note 3: Other operational details for automatic patrol tasks are the same as those for standard patrol tasks.

6. Selecting Task Mode

C -Lift Mode



note



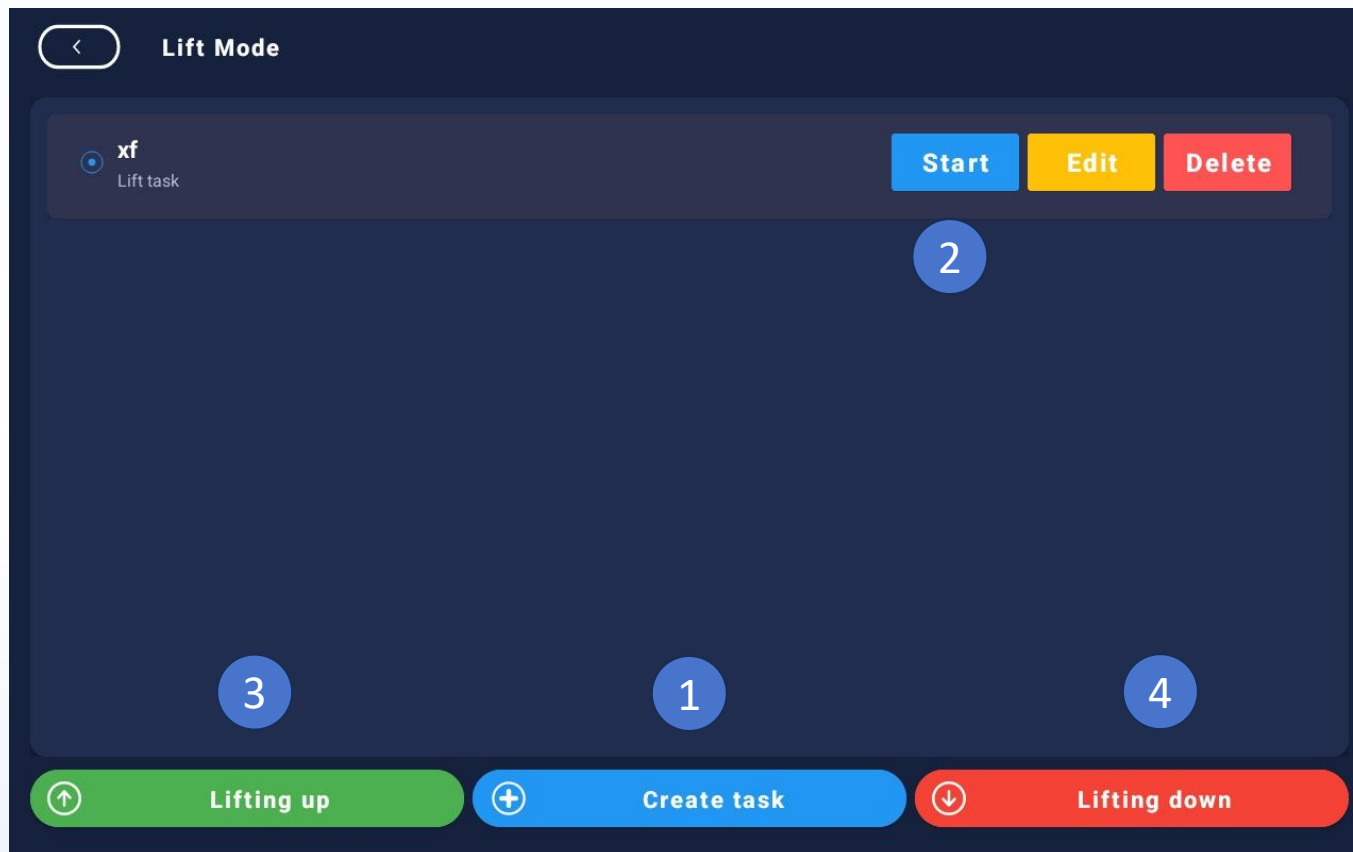
Operation Overview: On the main interface, tap "Lift Mode" to enter its interface and configure the AMR for point-to-point self lifting and transport.

6. Selecting Task Mode



note

C -Lift Mode



- ① Click "Create task", enter a task name. The created task will be displayed in the top panel.
- ② In the "Lift Mode" interface, click "Start" to initiate the task.
- ③ Click "Lifting up" to raise the AMR's lifting platform.
- ④ Click "Lifting down" to lower the AMR's lifting platform.

6. Selecting Task Mode

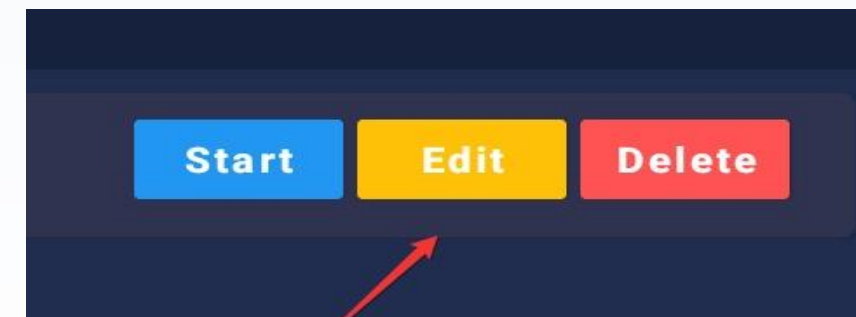
C - Lift Mode

Ensure the racks are correctly positioned at the designated lifting points prior to task execution.

The screenshot shows a 'Custom Tasks' interface. On the left, there is a list of tasks labeled A, B, C, D, and E. The main area displays the details for task A, which is highlighted with a red box. The details for task A include 'Execute Action: Recognizing lifting' and 'Stay Duration: 3'. Below this, task B is shown with 'Execute Action: Recognizing lowering' and 'Stay Duration: 3'. Task C is shown with 'Execute Action: Go out pod' and 'Stay Duration: 3'. At the bottom of the interface, there is a blue button labeled 'Save route'.



note



- ① To edit a task, click its "Edit" button, select locations, assign actions (Identify/Lift, Identify/Lower, Exit Location, Move), and set wait times.
- ② Click "Save route" to save the task configuration.

Simple Transport Task:

(1) Identify & Lift → (2) Identify & Lower → (3) Exit Storage Location

7. Main Interface Explanation

Other Parts of Main Interface

(1) Return to Charge

- ① When the AMR is idle or has no tasks, you can manually tap "Return to Standby Point". The AMR will then automatically navigate back to its designated standby location.
- ② "Special Locations" is the dedicated settings section for key points like charging stations and standby positions. The coordinates for the standby point are configured under "Location Settings" > "Special Locations".

(2) Return to Standby Point

- ① When the AMR is idle or has no tasks, you can manually tap "Return to Standby Point". The AMR will then automatically navigate back to its designated standby location.
- ② The coordinates for the standby point are configured under "Location Settings" > "Special Locations".

(3) AMR Status Bar

Status Bar Information:

- ① Operational Status: Idle, Running, Complete, Failed
- ② Exception Status: E-Stop Triggered, Bumper Activated, Navigation Laser Lost, Obstacle Detected, etc.
- ③ Mapping Quality: 0-100% (Higher values indicate more stable AMR operation)
- ④ Current Position: X, Y coordinates and Theta orientation angle

(4) Current Time

- ① Displays the year, month, day, and current real-time.
- ② Set the time for the current time zone in Screen Settings under 'Date & Time'.

8. Settings Overview

System Settings

From the Main Interface, tap "System Settings" to access the configuration and viewing menu for Location Settings, Basic Settings, Voice Settings, Map Settings, and About the Device.

(1) Location Settings

Location Settings include two functions: Working Locations and Special Locations.

(2) Basic Settings

- Volume & Speed: Adjust via sliders.
- Delivery Settings: Set dwell time.
- Language Settings: Select from available languages (Chinese/English/Korean/Japanese).
- Charging Settings: Enable/disable auto-charging & set low battery alert threshold.
- Password Verification: Enable/disable 4-digit integer password authentication.
- Restore Defaults: Clear all custom settings.
- Save Settings: Store custom configurations.



8. Settings Overview

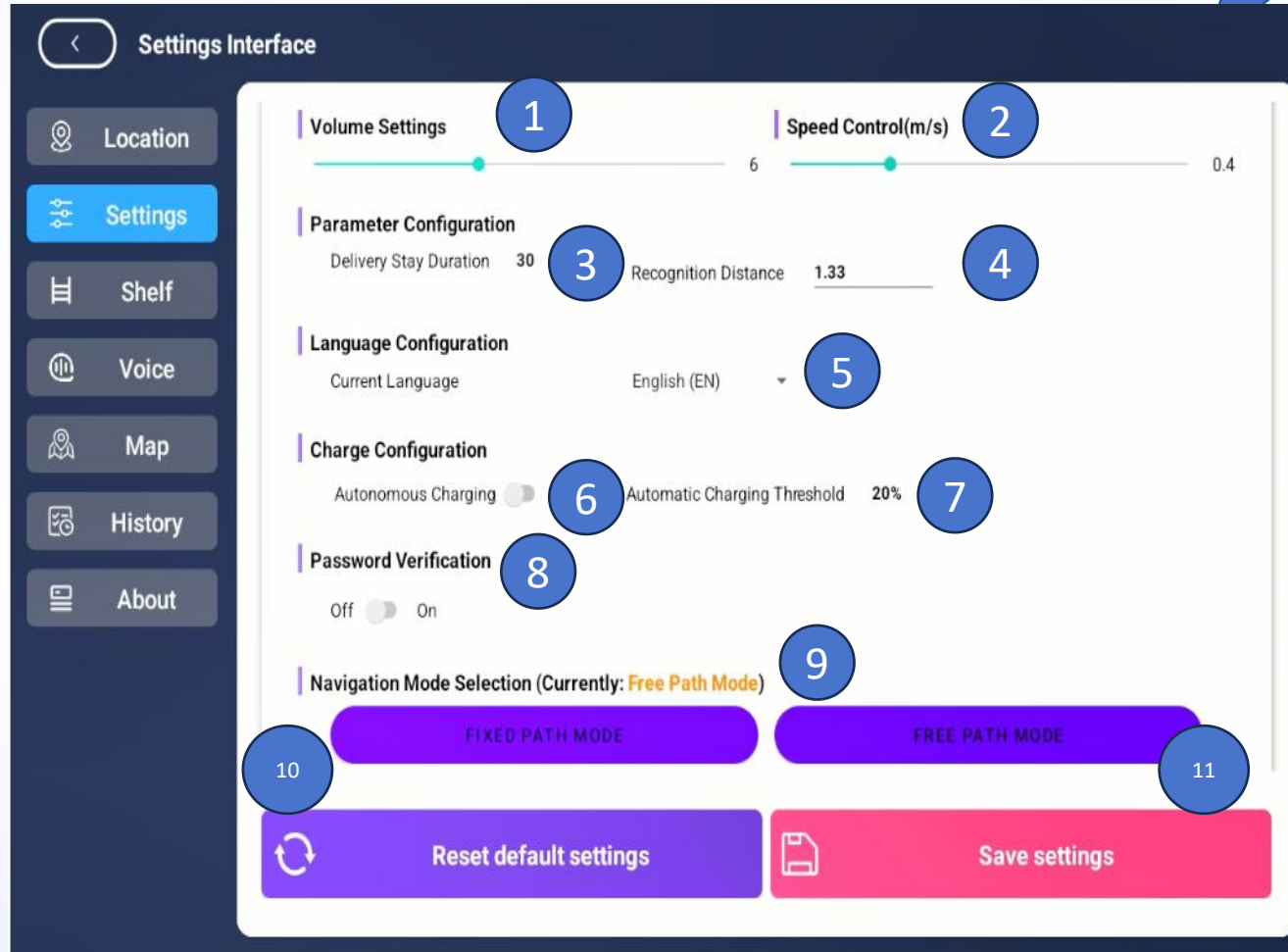
New settings must be saved to take effect; otherwise, changes will not be retained.



note

Ezhan

8-1. Settings Interface



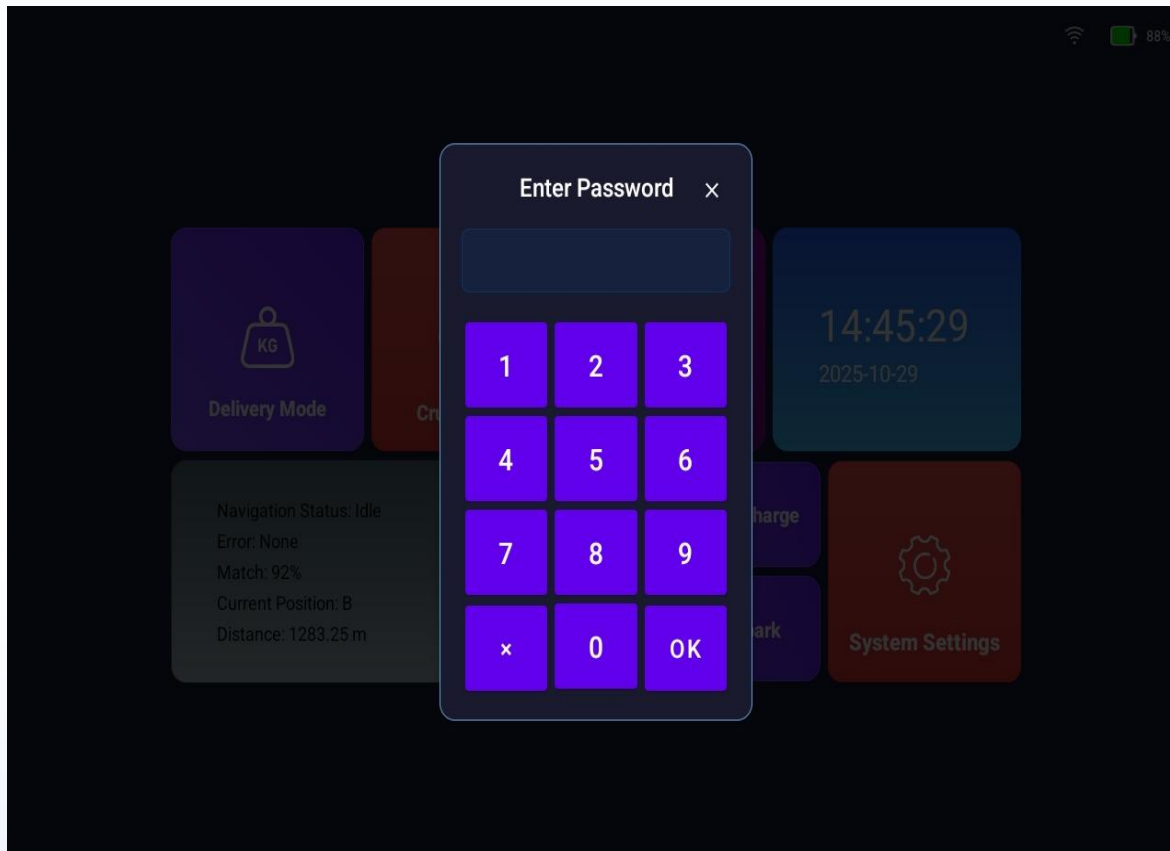
- ① Slide to adjust volume;
- ② Slide to adjust speed;
- ③ Set the AMR's waiting time at each station;
- ④ Set the distance between AMR and the material rack
- ⑤ Switch language;
- ⑥ Enable/Disable auto-charging;
- ⑦ Set AMR low battery alarm threshold;
- ⑧ Enable/Disable password verification;
- ⑨ Switch navigation mode
- ⑩ Restore default settings;
- ⑪ Save settings.

8. Settings Overview

8-1. Settings Interface –Password Configuration



note



- Note 1: New settings must be saved to take effect; otherwise changes will be discarded.
- Note 2: Click "Restore Default Settings" to reset to factory defaults.
- Note 3: Contact your supplier for password recovery if forgotten or lost.

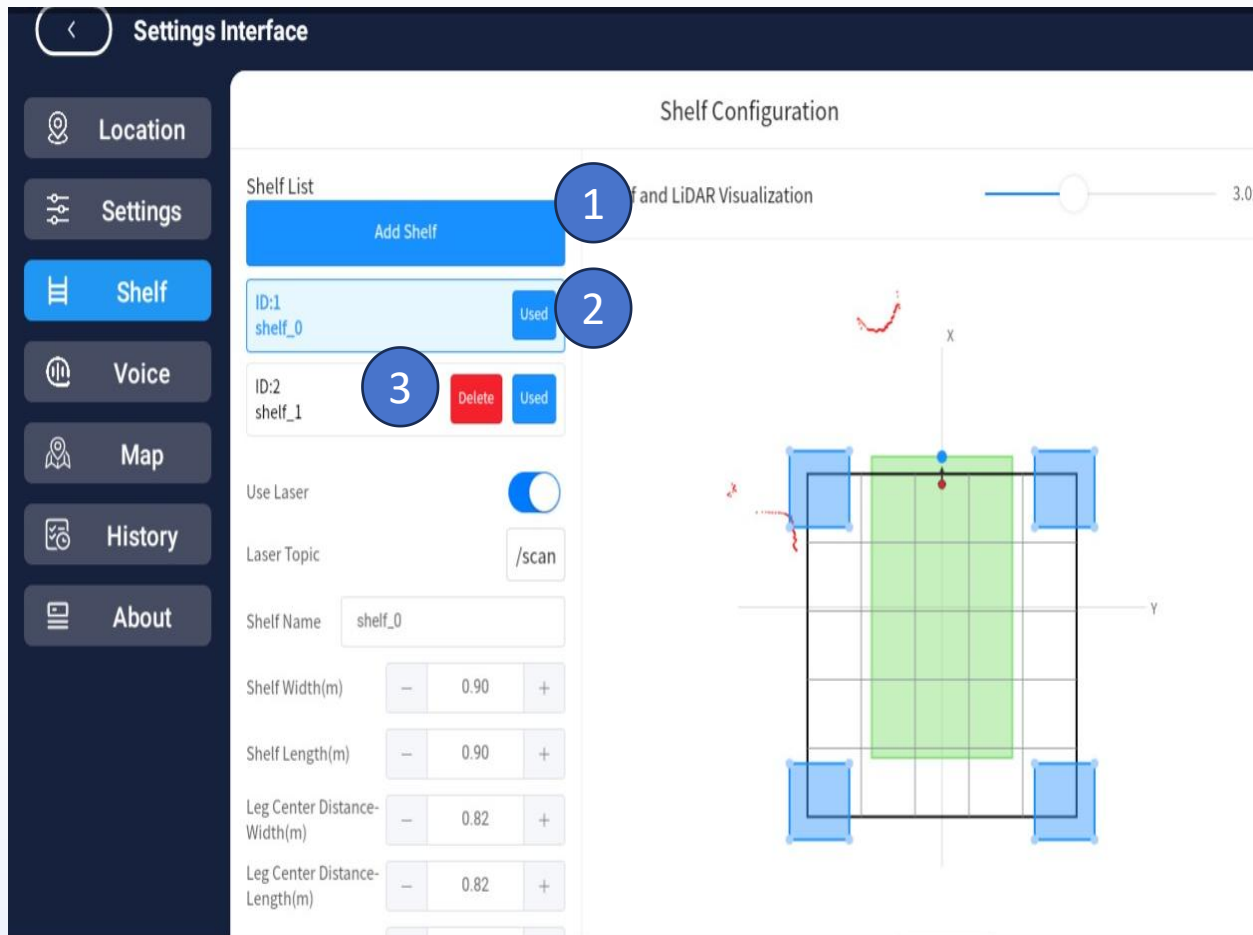


8. Settings Overview

8-2. Shelf Settings



note



Before moving the shelf, its parameters need to be configured.

- Note 1: Click "Add Shelf" to add a new shelf.
- Note 2: Click "Used" to indicate this shelf model is currently in use.
- Note 3: Should you wish to discard this shelf model, click "Delete".



8. Settings Overview

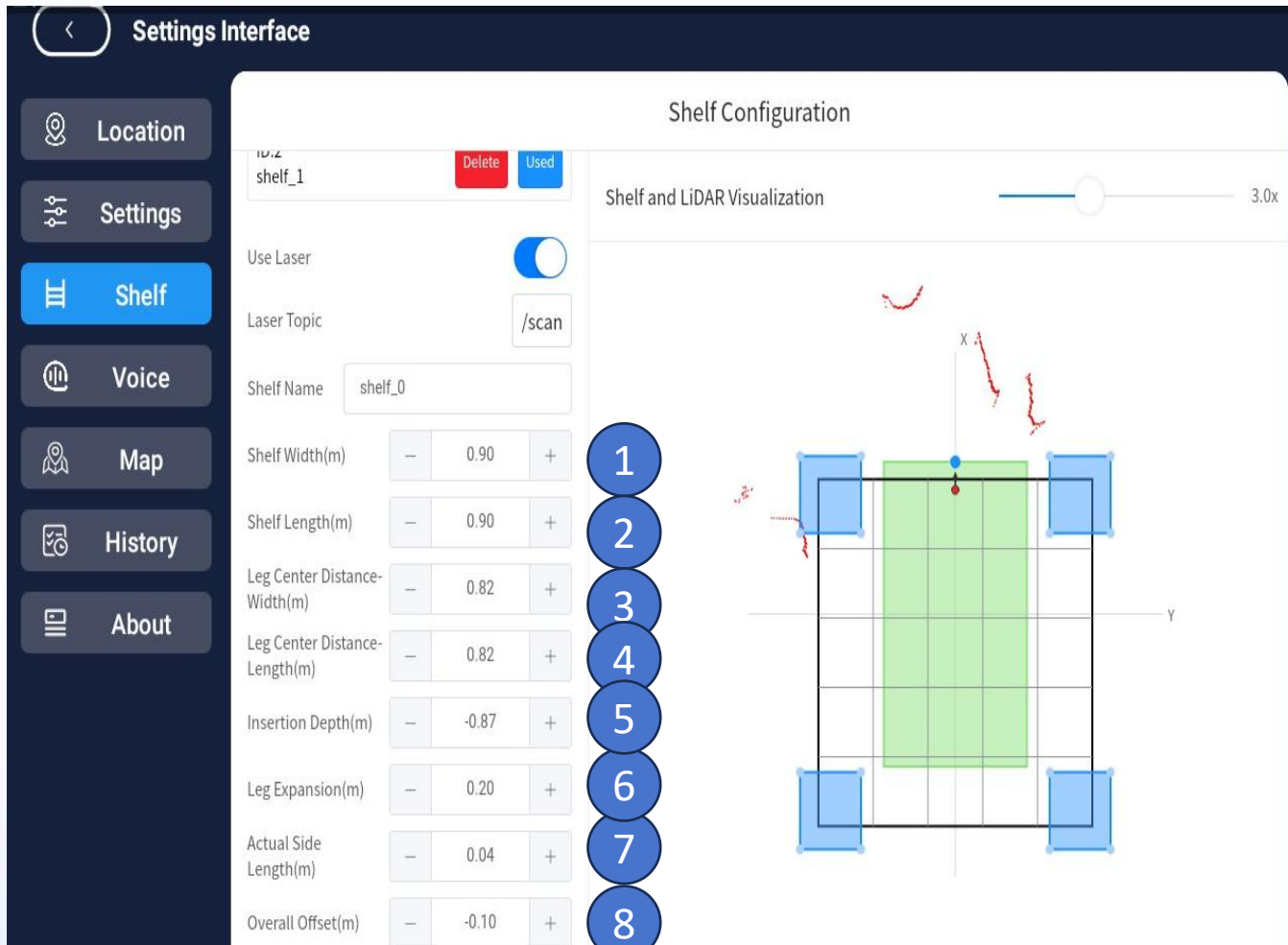
8-2. Shelf Settings



note

After clicking "Add Shelf", you need to configure some parameters for the shelf

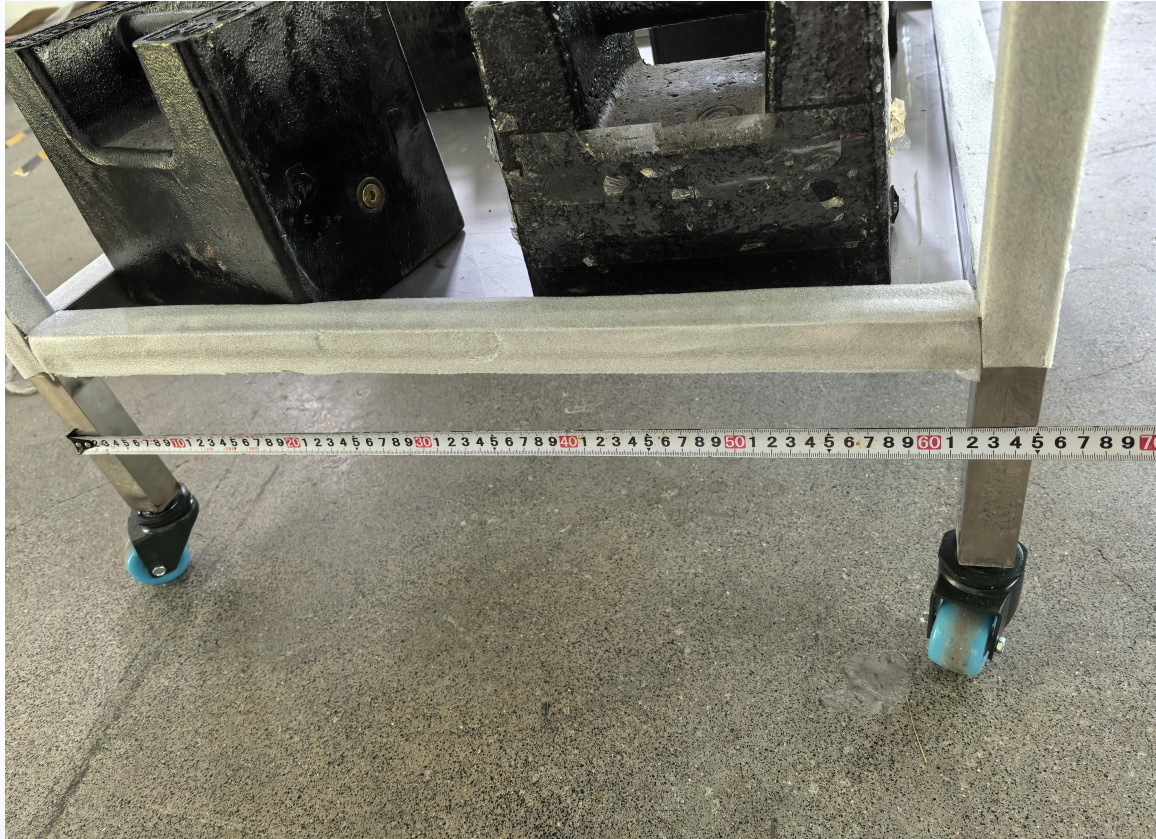
- ①: Enter the measured width of the shelf.
- ②: Enter the measured length of the shelf.
- ③: Set the Leg Center Distance (Width).
- ④: Set the Leg Center Distance (Length).
- ⑤: Set the Insertion Depth: -0.87m (default).
- ⑥: Set the leg Expansion: 0.2m (default).
- ⑦: Set Actual Leg Width: The physical width/thickness of an individual shelf leg.
- ⑧: Set Overall Offset: Adjust so the shelf leg (left image) is centered in the blue target square.



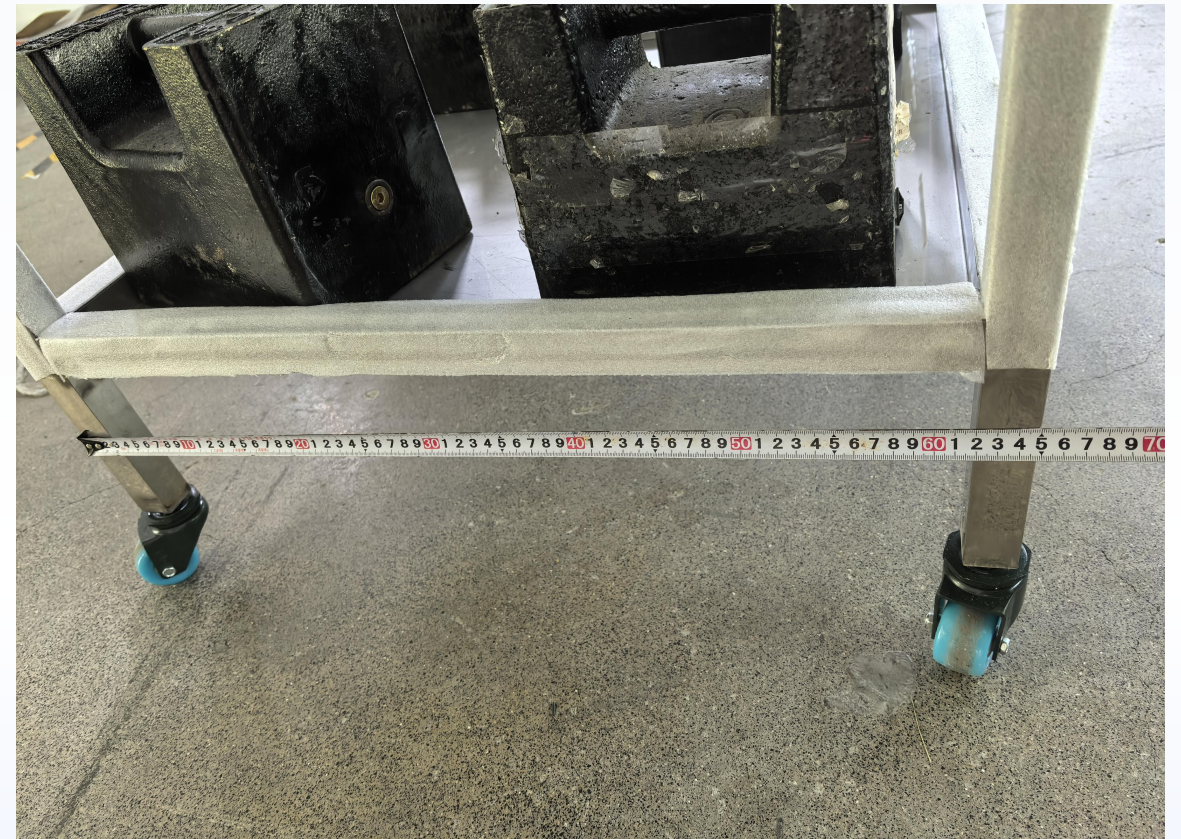
8. Settings Overview



8-2. Shelf Settings



Measure the shelf length



Measure the shelf width



8. Settings Overview

8-2. Shelf Settings



Measure the leg center distance width between the left and right shelf legs



Measure the leg center distance length between the front and rear shelf legs



Measure the actual leg width

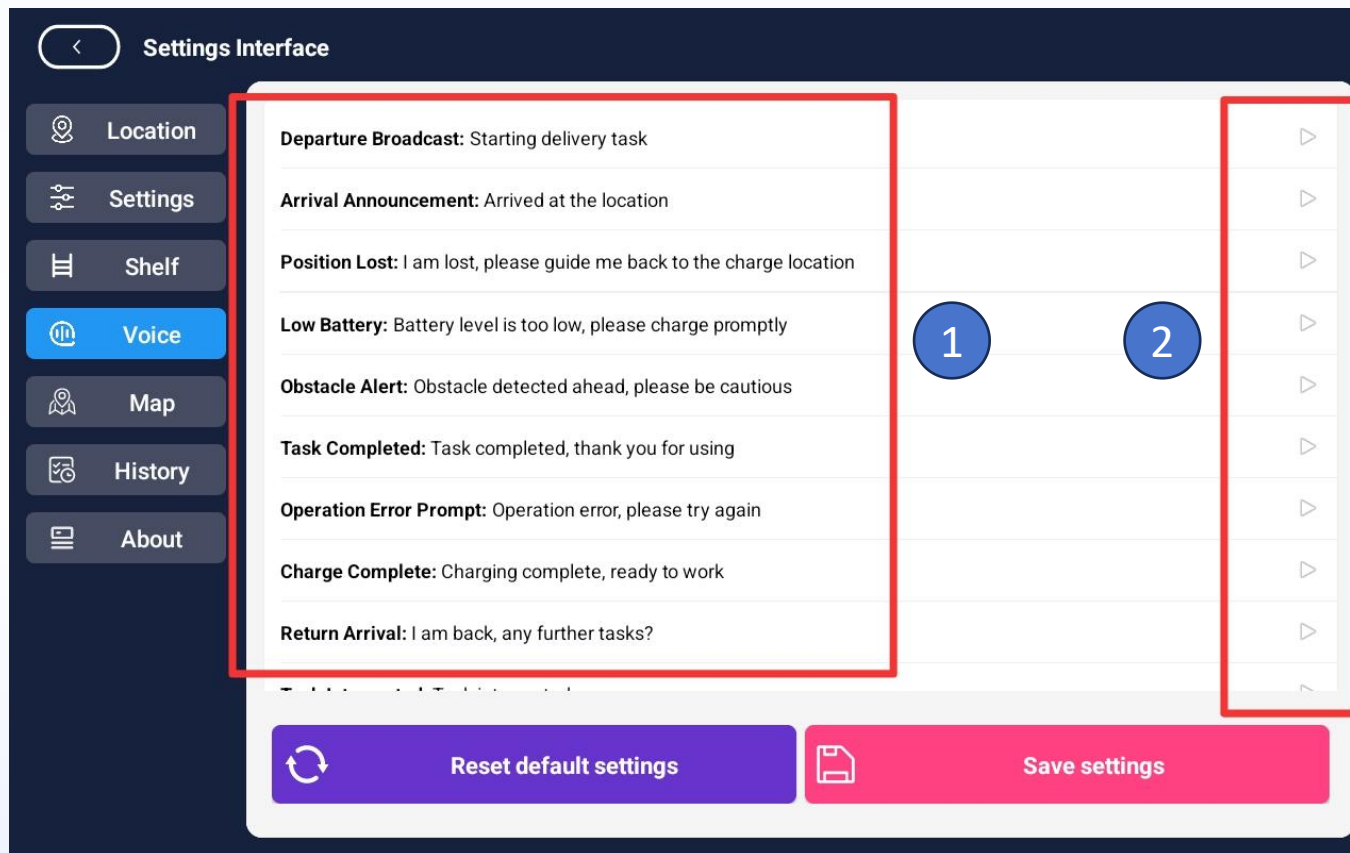


8. Settings Overview

8-3. Voice Settings



note



- ① Text for each voice prompt;
- ② Tap the input field to modify the prompt text;
- ③ Click "Play" to hear the corresponding voice prompt.

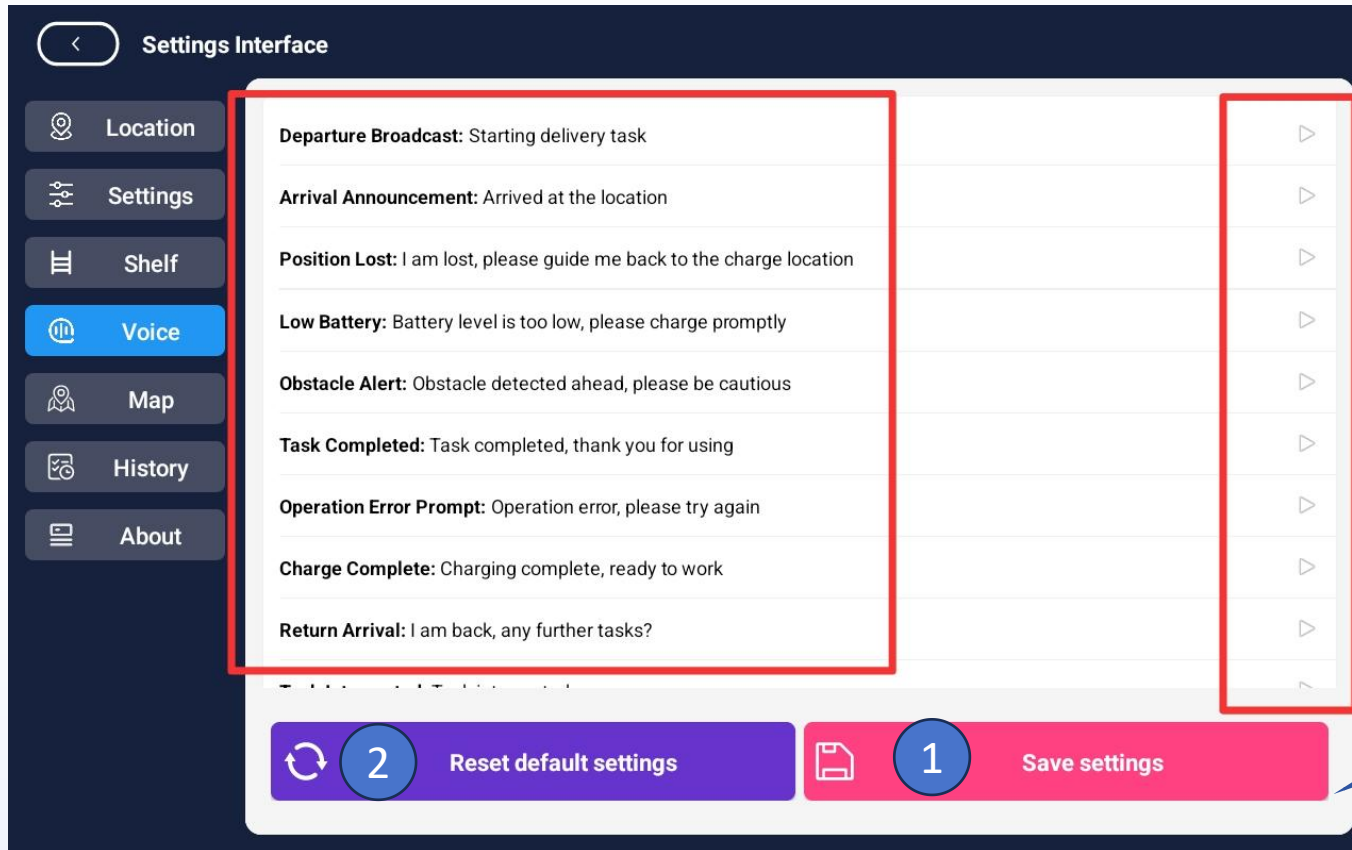


8. Settings Overview

8-3. Voice Settings



note



① Click "Save settings" to store voice prompt configurations.

② Click "Restore default settings" to reset to original defaults.

New settings must be saved to take effect; otherwise changes will not be retained.



8. Settings Overview

8-4. History Settings



note

Settings Interface

1 2 3 4 5

Start End Type Count Search

2026-02-02 10:37:46 All 3

Jack-up Task yy	Create Time: 2026-01-31 15:00	Duration: 00:00:00	Status: Canceled
Cruise Task 天天	Create Time: 2026-01-31 14:59	Duration: 00:00:03	Status: Canceled
Delivery Task 背负任务->工位2	Create Time: 2026-01-31 14:59	Duration: 00:00:16	Status: Canceled

- ① Task Start Time.
- ② Task End Time.
- ③ Task Types.
- ④ Task Counts.
- ⑤ Search: Find completed tasks based on different time settings and task types



9. Troubleshooting

Problem Type	Resolution
The robot's red light is lit.	Inspection Steps: Physical Vehicle Check: 1. Check if the bottom E-stop button is pressed. If pressed, rotate it to release. 2. Check if the pole-mounted E-stop button is pressed. If pressed, rotate it to release. 3. Check if the safety bumper is triggered. If triggered, press and then rotate the pole-mounted E-stop button to reset the system. Application Check: 1. If the AMR status bar shows "Soft E-stop triggered", press and then rotate the pole-mounted E-stop button to reset. 2. If the AMR status bar shows "Map initialization failed", navigate to Map Settings, re-select the target map, and enter it. 3. If the AMR status bar shows "Route timeout", access the map, set a start point in "Parameter Settings", then restart the AMR. After reboot, perform re-localization.
The remote control fails to command the robot.	1. Check that the white button on the remote was pressed to activate it; activation is confirmed by a green light. 2. Check if the operation mode selected is "Differential Drive".
Destination unreachable after setting waypoints.	1. Verify AMR Map Entry: Check if the AMR has entered the target map. If not, select the target map and tap "Enter Map", then refresh the target point locations. 2. Inspect Target Point Placement: Confirm whether the target point is placed on map noise points or within a virtual wall. If placed on noise, edit the map to erase the noise points. If placed inside a virtual wall, either recreate the point or reduce the virtual wall's effective range. 3. Validate Point Location Scope: Ensure the target point is within the target map's boundaries. If not, remap the area.
Robot not charging.	Check the Map: 1. Confirm whether the AMR has entered the target map. 2. Verify that a charging point exists on the map. 3. Check whether the localization quality is correct. Check the Charging Pile: 1. Check if the charging pile is correctly positioned. 2. Verify the charging interface height is properly aligned with the AMR's charging port.
Robot is silent.	1. In the app, navigate to System Settings > Basic Settings and check if the Volume Setting is set to 0. If it is, adjust the volume level as needed. 2. Under System Settings > Voice Settings, confirm that the voice broadcast language matches the current system language. If not, first click "Restore Default Settings", then click "Save Settings", and finally restart the app.

THANK YOU

E300 Series User Manual